CHAPTER 18

Ethers and Epoxides; Thiols and Sulfides



FIGURE 18.1 **The appalling and unforgettable odor of skunks is due to a mixture of several simple thiols**. (credit: modification of work "Striped Skunk" by Tom Friedel/Wikimedia Commons, CC BY 3.0)

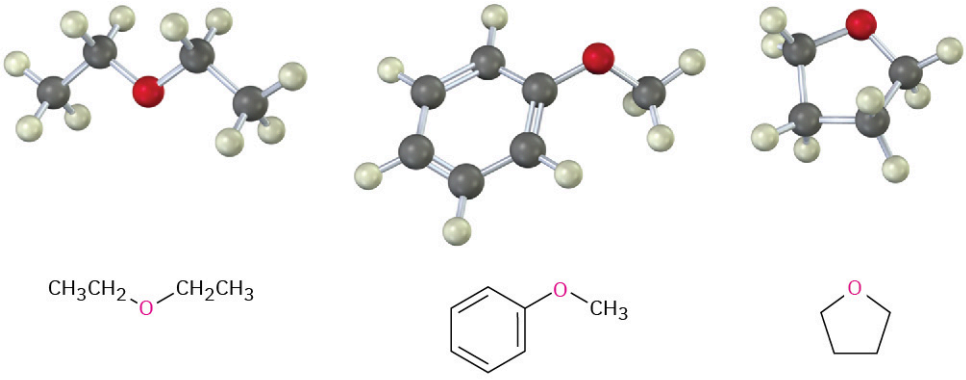
CHAPTER CONTENTS

18.1 Names and Properties of Ethers
18.2 Preparing Ethers
18.3 Reactions of Ethers: Acidic Cleavage
18.4 Cyclic Ethers: Epoxides
18.5 Reactions of Epoxides: Ring-Opening
18.6 Crown Ethers
18.7 Thiols and Sulfides
18.8 Spectroscopy of Ethers

WHY THIS CHAPTER? This chapter finishes the coverage of functional groups with C–O and C–S single bonds that was begun in the chapter on **Alcohols and Phenols**. We'll focus primarily on ethers and take only a brief look at thiols and sulfides before going on to an extensive coverage of compounds with C=O double bonds in Chapters 19 through 23.

Ethers (R–O–R′), like the alcohols we saw in the preceding chapter, are organic derivatives of water, but they have two organic groups bonded to the same oxygen atom rather than one. The organic groups might be alkyl, aryl, or vinylic, and the oxygen atom might be in an open chain or a ring.

Perhaps the most well-known ether is diethyl ether, which has a long history of medicinal use as an anesthetic and industrial use as a solvent. Other useful ethers include anisole, a pleasant-smelling aromatic ether used in perfumery, and tetrahydrofuran (THF), a cyclic ether often used as a solvent.

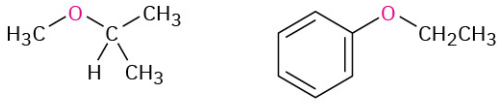


Diethyl ether **Anisole (methyl phenyl ether)** **Tetrahydrofuran**

Thiols (R–S–H) and sulfides (R–S–R′) are sulfur analogs of alcohols and ethers, respectively. Both functional groups are found in various biomolecules, although not as commonly as their oxygen-containing relatives.

18.1 Names and Properties of Ethers

Simple ethers with no other functional groups are named by identifying the two organic substituents and adding the word *ether*.



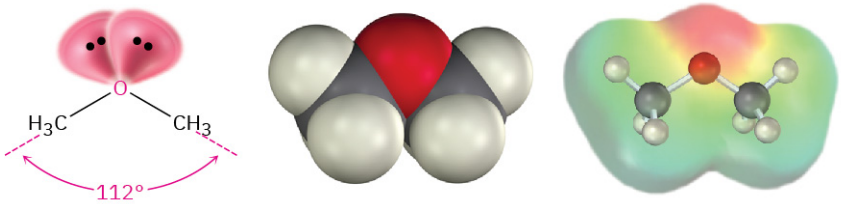
Isopropyl methyl ether **Ethyl phenyl ether**

If other functional groups are present, the ether part is considered an *alkoxy* substituent. For example:



***p*-Dimethoxybenzene** **4-*tert*-Butoxy-1-cyclohexene**

Like alcohols, ethers have nearly the same geometry as water. The R–O–R bonds have an approximately tetrahedral bond angle (112° in dimethyl ether), and the oxygen atom is sp^3-hybridized.

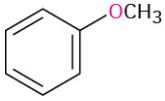
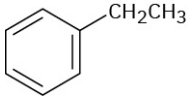


The electronegative oxygen atom gives ethers a slight dipole moment, and the boiling points of ethers are often slightly higher than the boiling points of comparable alkanes. TABLE 18.1 compares the boiling points of some common ethers and their corresponding hydrocarbons.

TABLE 18.1 Comparison of Boiling Points of Ethers and Hydrocarbons

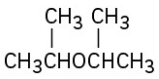
Ether	Boiling point °C	Hydrocarbon	Boiling point °C
CH_3OCH_3	−25	$CH_3CH_2CH_3$	−45
$CH_3CH_2OCH_2CH_3$	34.6	$CH_3CH_2CH_2CH_2CH_3$	36

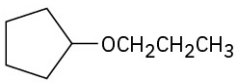
TABLE 18.1 Comparison of Boiling Points of Ethers and Hydrocarbons

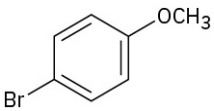
Ether	Boiling point °C	Hydrocarbon	Boiling point °C
	65		49
	158		136

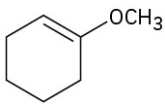
Ethers are relatively stable and unreactive in many respects, but some ethers react slowly with the oxygen in air to give *peroxides,* compounds that contain an O–O bond. The peroxides from low-molecular-weight ethers such as diisopropyl ether and tetrahydrofuran are explosive and extremely dangerous, even in small amounts. Ethers are very useful as solvents in the laboratory, but they must always be used cautiously and should not be stored for long periods of time.

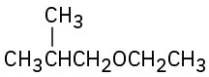
PROBLEM 18-1 Name the following ethers:

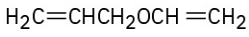
(a) 

(b) 

(c) 

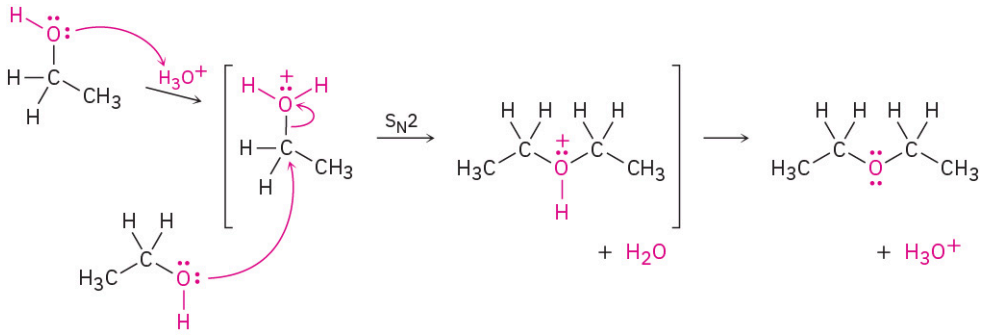
(d) 

(e) 

(f) 

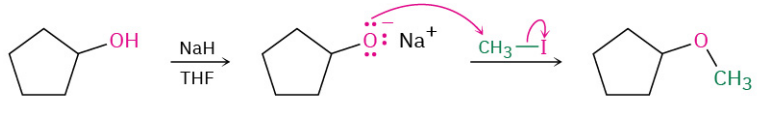
18.2 Preparing Ethers

Diethyl ether and other simple symmetrical ethers are prepared industrially by the sulfuric-acid-catalyzed reaction of alcohols. The reaction occurs by S_N2 displacement of water from a protonated ethanol molecule by the oxygen atom of a second ethanol. Unfortunately, this method is limited to use with primary alcohols because secondary and tertiary alcohols dehydrate by an E1 mechanism to yield alkenes (Section 17.6).



The Williamson Ether Synthesis

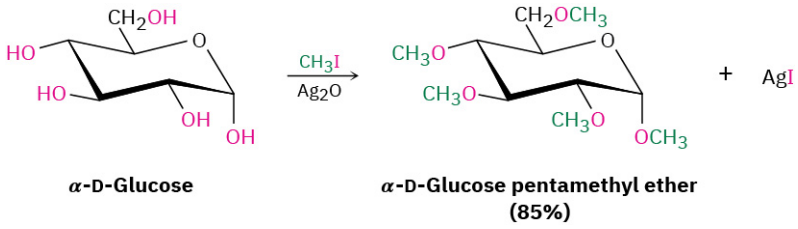
The most generally useful method of preparing ethers is the **Williamson ether synthesis**, in which an alkoxide ion reacts with a primary alkyl halide or tosylate in an S_N2 reaction. As we saw in Section 17.2, the alkoxide ion is normally prepared by reaction of an alcohol with a strong base such as sodium hydride, NaH.



Cyclopentanol **Alkoxide ion** **Cyclopentyl methyl ether (74%)**

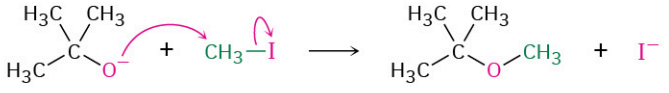
A useful variation of the Williamson synthesis involves silver oxide, Ag_2O, as a mild base rather than NaH. Under these conditions, the free alcohol reacts directly with the alkyl halide, so there is no need to preform the metal

alkoxide intermediate. Sugars react particularly well; glucose, for example, reacts with excess iodomethane in the presence of Ag_2O to generate a pentaether in 85% yield.

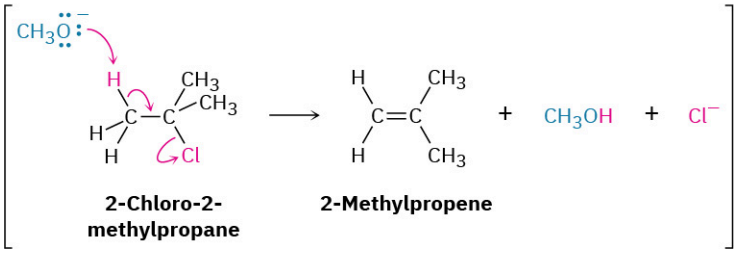


α-D-Glucose **α-D-Glucose pentamethyl ether (85%)**

Because the Williamson synthesis is an S_N2 reaction, it is subject to all the usual constraints, as discussed in **Section 11.3**. Primary halides and tosylates work best because competitive E2 elimination can occur with more hindered substrates. Unsymmetrical ethers should therefore be synthesized by reaction between the more hindered alkoxide partner and less hindered halide partner rather than vice versa. For example, *tert*-butyl methyl ether, a substance used in the 1990s as an octane booster in gasoline, is best prepared by reaction of *tert*-butoxide ion with 2-chloro-2-methylpropane.



***tert*-Butoxide** **Iodomethane** ***tert*-Butyl methyl ether**



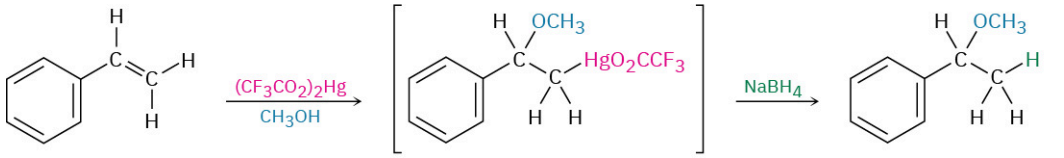
2-Chloro-2-methylpropane **2-Methylpropene**

PROBLEM 18-2 Why do you suppose only symmetrical ethers are prepared by the sulfuric-acid-catalyzed dehydration procedure? What product(s) would you expect if ethanol and 1-propanol were allowed to react together? In what ratio would the products be formed if the two alcohols were of equal reactivity?

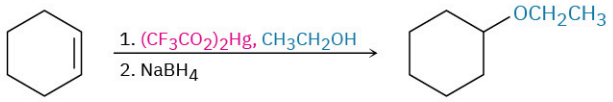
PROBLEM 18-3 How would you prepare the following ethers using a Williamson synthesis?
(a) Methyl propyl ether **(b)** Anisole (methyl phenyl ether) **(c)** Benzyl isopropyl ether
(d) Ethyl 2,2-dimethylpropyl ether

Alkoxymercuration of Alkenes

We saw in **Section 8.4** that alkenes react with water in the presence of mercuric acetate to yield a hydroxymercuration product. Subsequent treatment with $NaBH_4$ breaks the C–Hg bond and yields the alcohol. A similar **alkoxymercuration** reaction occurs when an alkene is treated with an alcohol in the presence of mercuric acetate or, even better, mercuric trifluoroacetate, $(CF_3CO_2)_2Hg$. Demercuration by reaction with $NaBH_4$ then yields an ether. The net result is Markovnikov addition of the alcohol to the alkene.



Styrene **1-Methoxy-1-phenylethane (97%)**



Cyclohexene **Cyclohexyl ethyl ether (100%)**

The mechanism of the alkoxymercuration reaction is similar to that described in **Section 8.4** for hydroxymercuration. The reaction is initiated by electrophilic addition of Hg^{2+} to the alkene, followed by reaction of the intermediate cation with alcohol and reduction of the C–Hg bond by $NaBH_4$. A variety of alcohols and alkenes can be used in alkoxymercuration. Primary, secondary, and even tertiary alcohols react well, but ditertiary ethers can't be prepared because of steric hindrance.

WORKED EXAMPLE 18.1

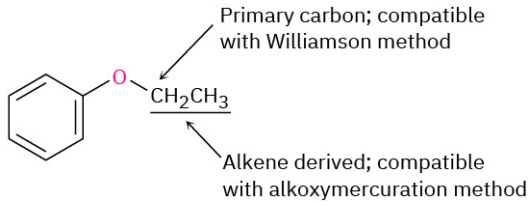
Synthesizing an Ether

How would you prepare ethyl phenyl ether? Use whichever method you think is more appropriate, Williamson synthesis or the alkoxymercuration reaction.

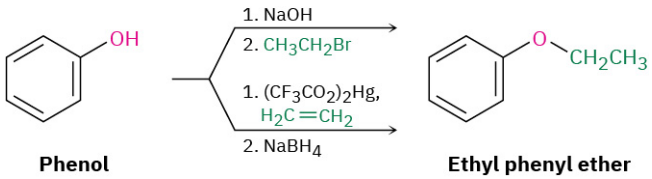
Strategy

Draw the target ether, identify the two groups attached to oxygen, and recall the limitations of the two methods for preparing ethers. Williamson synthesis uses an S_N2 reaction and requires that one of the two groups attached to oxygen be either secondary or (preferably) primary. The alkoxymercuration reaction requires that one of the two groups come from an alkene precursor. Ethyl phenyl ether could be made by either method.

Ethyl phenyl ether



Solution



Phenol **Ethyl phenyl ether**

PROBLEM 18-4 Review the mechanism of oxymercuration shown in Figure 8.4, and then write the mechanism of the alkoxymercuration reaction of 1-methylcyclopentene with ethanol. Use curved arrows to show the electron flow in each step.

PROBLEM 18-5 How would you prepare the following ethers? Use whichever method you think is more appropriate, Williamson synthesis or the alkoxymercuration reaction.

(a) Butyl cyclohexyl ether **(b)** Benzyl ethyl ether ($C_6H_5CH_2OCH_2CH_3$)

(c) *sec*-Butyl *tert*-butyl ether **(d)** Tetrahydrofuran

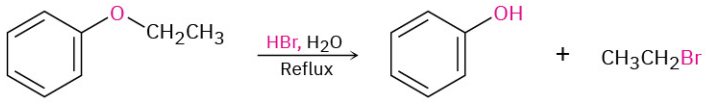
PROBLEM 18-6 Rank the following halides in order of their reactivity in the Williamson synthesis:

(a) Bromoethane, 2-bromopropane, bromobenzene

(b) Chloroethane, bromoethane, 1-Iodopropene

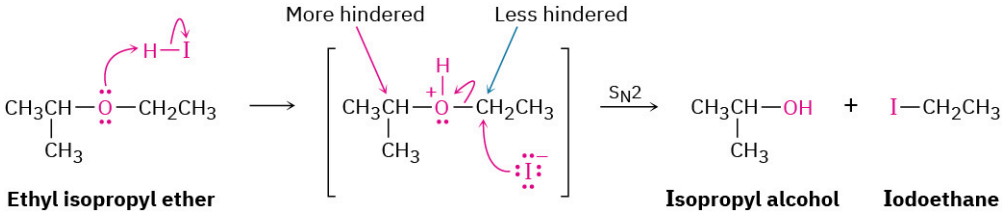
18.3 Reactions of Ethers: Acidic Cleavage

Ethers are unreactive to many reagents used in organic chemistry, a property that accounts for their wide use as reaction solvents. Halogens, dilute acids, bases, and nucleophiles have no effect on most ethers. In fact, ethers undergo only one truly general reaction—they are cleaved by strong acids. Aqueous HBr and HI both work well, but HCl does not cleave ethers.



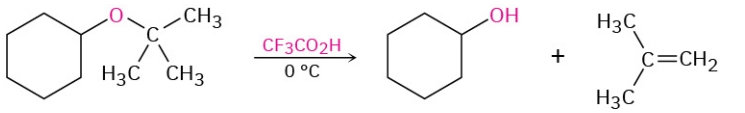
Ethyl phenyl ether **Phenol** **Bromoethane**

Acidic ether cleavages are typical nucleophilic substitution reactions and take place by either S_N1 or S_N2 mechanisms, depending on the structure of the substrate. Ethers with only primary and secondary alkyl groups react by an S_N2 mechanism, in which I^- or Br^- reacts with the protonated ether at the less hindered site. This usually results in a selective cleavage into a single alcohol and a single alkyl halide. For example, ethyl isopropyl ether exclusively yields isopropyl alcohol and iodoethane on cleavage by HI because nucleophilic attack by iodide ion occurs at the less hindered primary site rather than at the more hindered secondary site.



Ethyl isopropyl ether **Isopropyl alcohol** **Iodoethane**

Ethers with a tertiary, benzylic, or allylic group cleave by either an S_N1 or E1 mechanism because these substrates can produce stable intermediate carbocations. These reactions are often fast and take place at moderate temperatures. *tert*-Butyl ethers, for example, react by an E1 mechanism on treatment with trifluoroacetic acid at 0 °C. We'll see in **Section 26.7** that this reaction is often used in the laboratory synthesis of peptides.

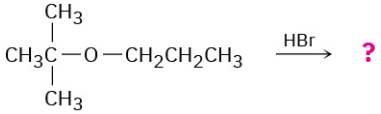


***tert*-Butyl cyclohexyl ether** **Cyclohexanol (90%)** **2-Methylpropene**

WORKED EXAMPLE 18.2

Predicting the Product of an Ether Cleavage Reaction

Predict the products of the following reaction:

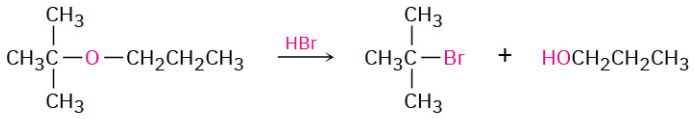


Strategy

Identify the substitution pattern of the two groups attached to oxygen—in this case a tertiary alkyl group and a primary alkyl group. Then recall the guidelines for ether cleavages. An ether with only primary and secondary alkyl groups usually undergoes cleavage by S_N2 attack of a nucleophile on the less hindered alkyl group, but an ether with a tertiary alkyl group usually undergoes cleavage by an S_N1 mechanism. In this case, an S_N1 cleavage

of the tertiary C–O bond will occur, giving 1-propanol and a tertiary alkyl bromide. In addition, a competitive E1 reaction leading to alkene might occur.

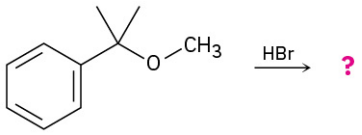
Solution



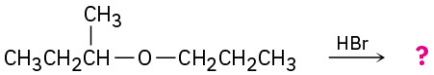
***tert*-Butyl propyl ether** **2-Bromo-2-methylpropane** **1-Propanol**

PROBLEM 18-7 Predict the products of the following reactions:

(a)



(b)

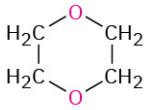


PROBLEM 18-8 Write the mechanism of the acid-induced cleavage of *tert*-butyl cyclohexyl ether to yield cyclohexanol and 2-methylpropene.

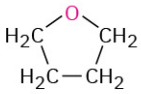
PROBLEM 18-9 Why are HI and HBr more effective than HCl in cleaving ethers? (See **Section 11.3**.)

18.4 Cyclic Ethers: Epoxides

For the most part, cyclic ethers behave like acyclic ethers. The chemistry of the ether functional group is the same, whether it's in an open chain or in a ring. Common cyclic ethers such as tetrahydrofuran and dioxane are often used as solvents because of their inertness, yet they can be cleaved by strong acids.



1,4-Dioxane

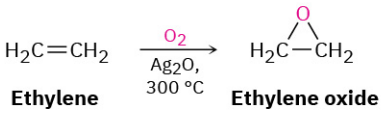


Tetrahydrofuran

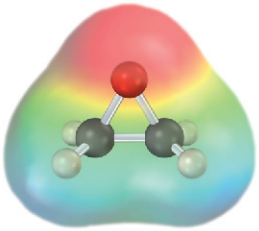
The one group of cyclic ethers that behaves differently from open-chain ethers are the three-membered-ring compounds called *epoxides,* or *oxiranes,* which we saw in Section 8.7. The strain of the three-membered ring gives epoxides unique chemical reactivity.

Ethylene oxide, the simplest epoxide, is an intermediate in the manufacture of both ethylene glycol, used for automobile antifreeze, and polyester polymers. Approximately 37 million tons of ethylene oxide is produced worldwide each year, most of it by air oxidation of ethylene over a silver oxide catalyst at 300 °C. This process is not useful for other epoxides, however, and is of little value in the laboratory.

Note that the name *ethylene oxide* is not a systematic one because the *-ene* ending implies the presence of a double bond in the molecule. The name is frequently used, however, because ethylene oxide is derived *from* ethylene by addition of an oxygen atom. Other simple epoxides are named similarly. The systematic name for ethylene oxide is 1,2-epoxyethane.

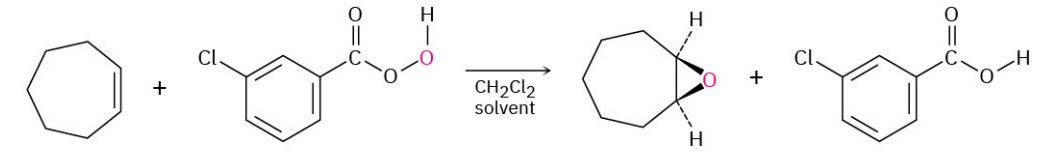


Ethylene **Ethylene oxide**

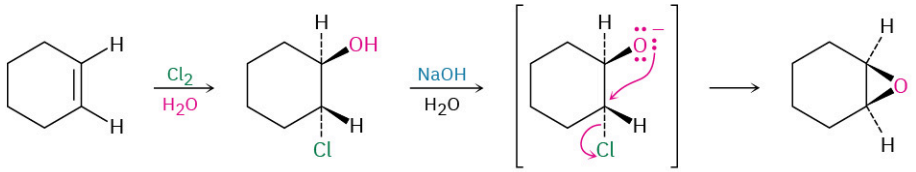


In the laboratory, as we saw in Section 8.7, epoxides are usually prepared by treatment of an alkene with a

peroxyacid (RCO_3H), typically *m*-chloroperoxybenzoic acid.



Epoxides can also be prepared from halohydrins, themselves produced by electrophilic addition of Cl_2 or Br2 to an alkene in the presence of water (Section 8.3). When a halohydrin is treated with base, HX is eliminated and an epoxide is produced by an intramolecular Williamson ether synthesis. That is, the nucleophilic alkoxide ion and the electrophilic alkyl halide are in the same molecule.



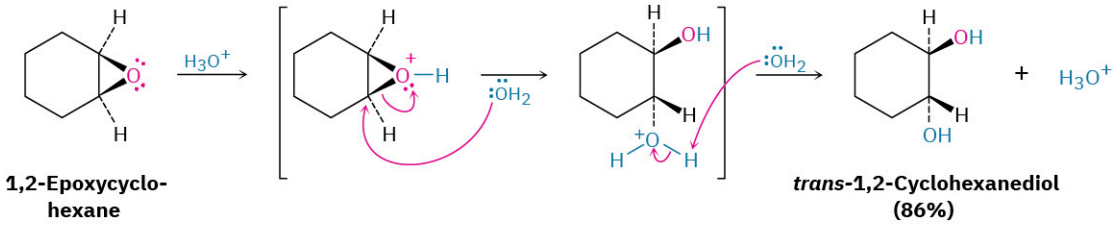
PROBLEM 18-10 What product would you expect from reaction of *trans*-2-butene with *m*-chloroperoxybenzoic acid? Show the stereochemistry.

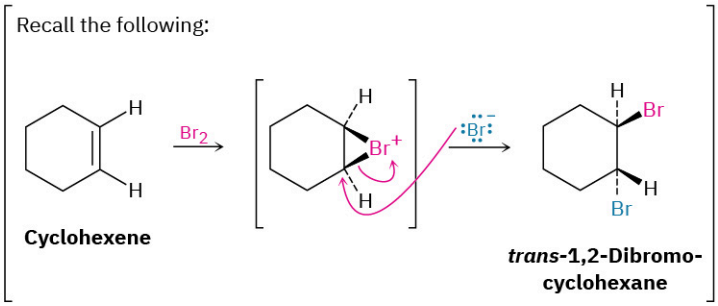
PROBLEM 18-11 Reaction of *cis*-2-butene with *m*-chloroperoxybenzoic acid yields an epoxide different from that obtained by reaction of the trans isomer. Explain.

18.5 Reactions of Epoxides: Ring-Opening

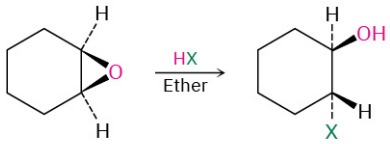
Acid-Catalyzed Epoxide Opening

Epoxides are cleaved by treatment with acid just as other ethers are, but under much milder conditions because of ring strain. As we saw in Section 8.7, dilute aqueous acid at room temperature is sufficient for facilitating the hydrolysis of epoxides to give 1,2-diols, also called *vicinal glycols.* (The word *vicinal* means "adjacent," and a *glycol* is a diol.) The epoxide cleavage takes place by S_N2-like backside attack of a nucleophile on the protonated epoxide, giving a trans-1,2-diol as product.





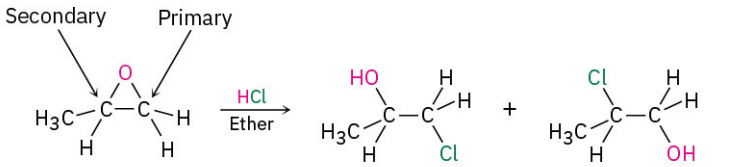
Epoxides can also be opened by reaction with acids other than H_3O^+. If anhydrous HX is used, for instance, an epoxide is converted into a trans halohydrin.



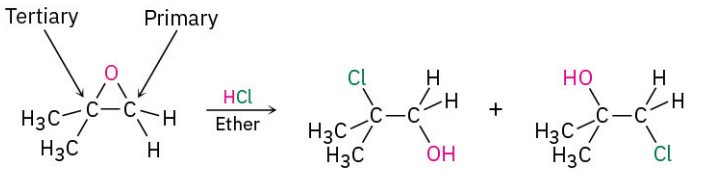
A trans 2-halocyclohexanol

where X = F, Br, Cl, or I

The regiochemistry of acid-catalyzed ring-opening depends on the epoxide's structure, and a mixture of products is often formed. When both epoxide carbon atoms are either primary or secondary, attack of the nucleophile occurs primarily at the less highly substituted site: an S_N2-like result. When one of the epoxide carbon atoms is tertiary, however, nucleophilic attack occurs primarily at the *more* highly substituted site—an S_N1-like result. Thus, 1,2-epoxypropane reacts with HCl to give primarily 1-chloro-2-propanol, but 2-methyl-1,2-epoxypropane gives 2-chloro-2-methyl-1- propanol as the major product.



1,2-Epoxypropane **1-Chloro-2-propanol (90%)** **2-Chloro-1-propanol (10%)**



2-Methyl-1,2-epoxypropane **2-Chloro-2-methyl-1-propanol (60%)** **1-Chloro-2-methyl-2-propanol (40%)**

The mechanisms of these acid-catalyzed epoxide openings are more complex than they initially appear. They seem to be neither purely S_N1 nor S_N2 but instead to be midway between the two extremes and to have characteristics of both. For instance, take the reaction of 1,2-epoxy-1-methylcyclohexane with HBr, shown in **FIGURE 18.2**. The reaction yields only a single stereoisomer of 2-bromo-2-methylcyclohexanol in which the –Br and –OH groups are trans, an S_N2-like result caused by backside displacement of the epoxide oxygen. But the fact that Br^- attacks the more hindered tertiary side of the epoxide rather than the less hindered secondary side is an S_N1-like result in which the more stable, tertiary carbocation is involved.

Evidently, the transition state for acid-catalyzed epoxide opening has an S_N2-like geometry but also has a high degree of S_N1-like carbocationic character. Because the positive charge in the protonated epoxide is shared by the more highly substituted carbon atom, backside attack of Br^- occurs at the more highly substituted site.

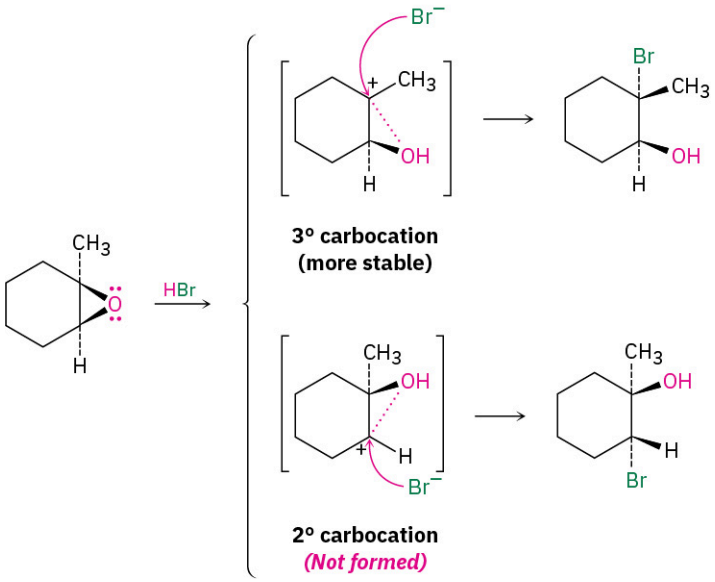
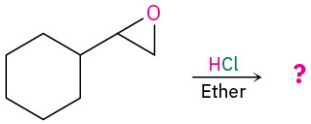


FIGURE 18.2 **Ring-opening of 1,2-epoxy-1-methylcyclohexane with HBr**. There is a high degree of S_N1-like carbocation character in the transition state, which leads to backside attack of the nucleophile at the tertiary center and to formation of a product isomer that has –Br and –OH groups trans.

WORKED EXAMPLE 18.3

Predicting the Product of Epoxide Ring-Opening

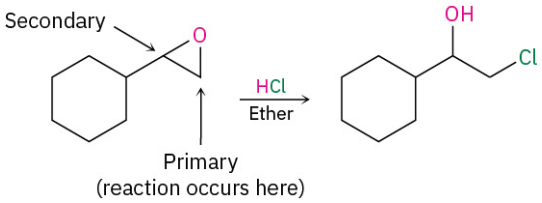
Predict the major product of the following reaction:



Strategy

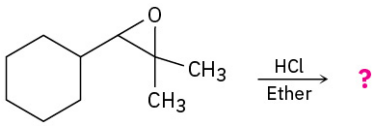
Identify the substitution pattern of the two epoxide carbon atoms. In this case, one carbon is secondary and one is primary. Then recall the guidelines for epoxide cleavages. An epoxide with only primary and secondary carbons usually undergoes cleavage by S_N2-like attack of a nucleophile on the less hindered carbon, but an epoxide with a tertiary carbon atom usually undergoes cleavage by backside attack on the more hindered carbon. In this case, an S_N2 cleavage of the primary C–O epoxide bond will occur.

Solution

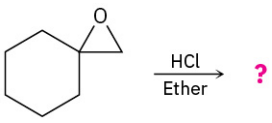


PROBLEM 18-12 Predict the major product of each of the following reactions:

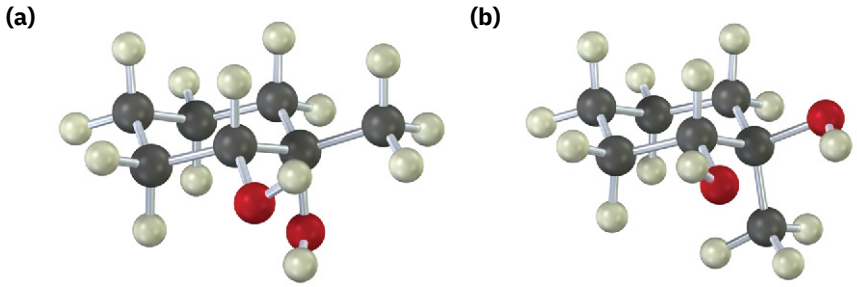
(a)



(b)

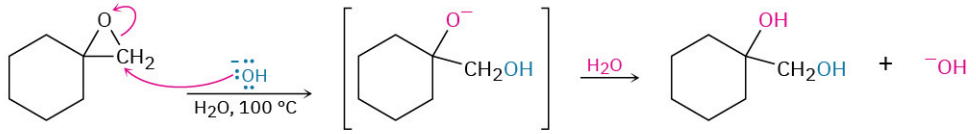


PROBLEM 18-13 How would you prepare the following diols?



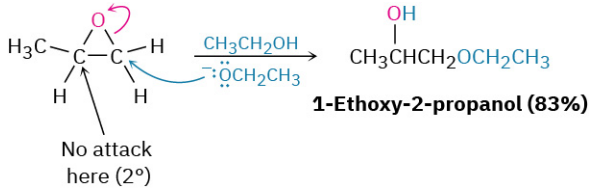
Base-Catalyzed Epoxide Opening

Unlike other ethers, epoxide rings can be cleaved by bases and nucleophiles as well as by acid. Although an ether oxygen is normally a poor leaving group in an S_N2 reaction (**Section 11.3**), the strain of the three-membered ring causes epoxides to react with hydroxide ion at elevated temperatures.



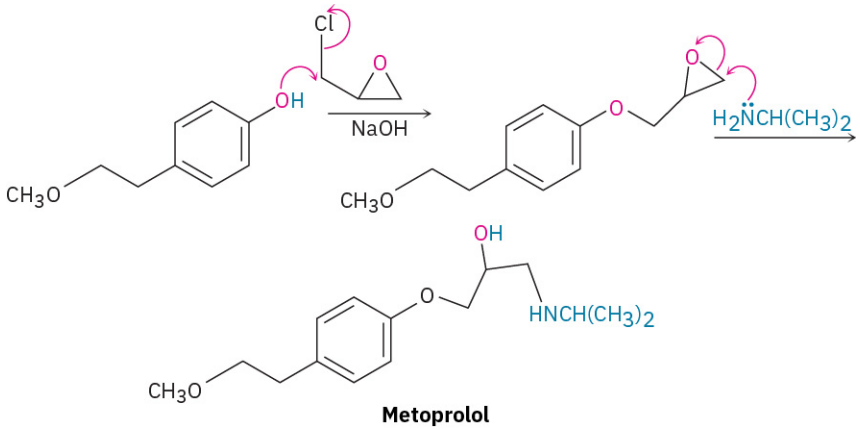
Methylenecyclohexane oxide **1-Hydroxymethyl-cyclohexanol (70%)**

Base-catalyzed epoxide opening is a typical S_N2 reaction in which attack of the nucleophile takes place at the less hindered epoxide carbon. For example, 1,2-epoxypropane reacts with ethoxide ion exclusively at the less highly substituted, primary carbon to give 1-ethoxy-2-propanol.



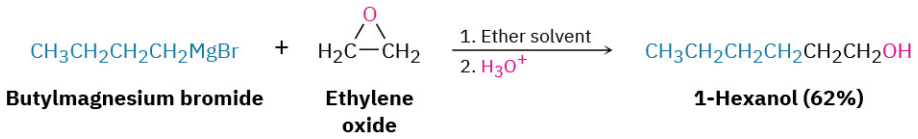
1-Ethoxy-2-propanol (83%)

Many different nucleophiles can be used for epoxide opening, including amines (RNH_2 or R_2NH) and Grignard reagents (RMgX). An example of an amine reacting with an epoxide occurs in the commercial synthesis of metoprolol, a so-called β-blocker that is used for treatment of cardiac arrhythmias, hypertension, and heart attacks. β-Blockers are among the most widely prescribed drugs in the world.



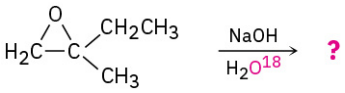
Metoprolol

A similar nucleophilic ring-opening occurs when epoxides are treated with Grignard reagents. Ethylene oxide is frequently used, thereby allowing the conversion of a Grignard reagent into a primary alcohol having two more carbons than the starting alkyl halide. 1-Bromobutane, for example, is converted into 1-hexanol by reaction of its Grignard reagent with ethylene oxide.

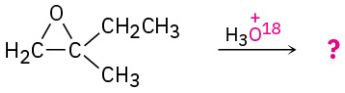


PROBLEM 18-14 Predict the major product of the following reactions:

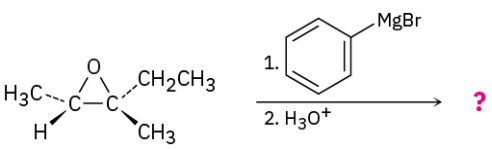
(a)



(b)

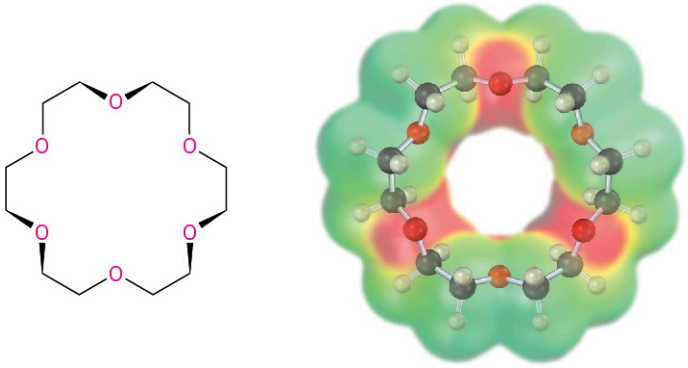


(c)



18.6 Crown Ethers

Crown ethers, discovered in the early 1960s by Charles Pedersen at the DuPont Company, are a relatively recent addition to the ether family. They are named according to the general format *x*-crown-*y*, where *x* is the total number of atoms in the ring and *y* is the number of oxygen atoms. Thus, 18-crown-6 ether is an 18-membered ring containing 6 ether oxygen atoms. Note the size and negative (red) character of the crown ether cavity in the following electrostatic potential map.

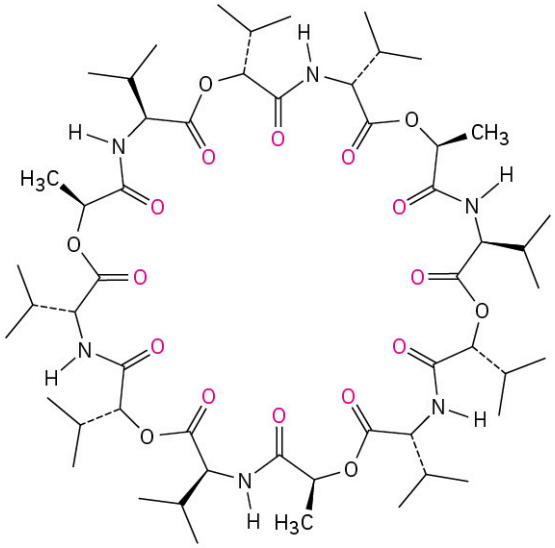


18-Crown-6 ether

The importance of crown ethers stems from their ability to sequester specific metal cations in the center of the polyether cavity. 18-Crown-6, for example, binds strongly with potassium ion. As a result, a solution of 18-crown-6 in a nonpolar organic solvent will dissolve many potassium salts. Potassium permanganate, $KMnO_4$, dissolves in toluene in the presence of 18-crown-6, for instance, and the resulting solution is a valuable reagent for oxidizing alkenes.

The effect of using a crown ether to dissolve an inorganic salt in a hydrocarbon or ether solvent is similar to the effect of dissolving the salt in a polar aprotic solvent such as DMSO, DMF, or HMPA (**Section 11.3**). In both cases, the metal cation is strongly solvated, leaving the anion bare. Thus, the S_N2 reactivity of an anion is tremendously enhanced in the presence of a crown ether.

Although crown ethers do not occur naturally, a group of compounds called *ionophores* have similar ion-binding properties. Produced by various microorganisms, ionophores are fat-soluble molecules that bind to specific ions and facilitate transport of the ions through biological membranes. The antibiotic valinomycin, for instance, binds specifically to K^+ ions with a ten-thousandfold selectivity over Na^+.



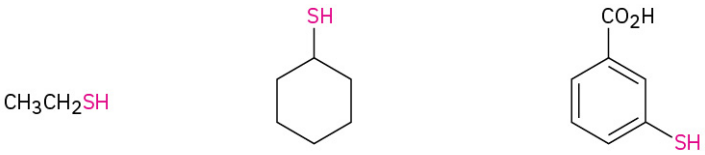
Valinomycin

PROBLEM 18-15 15-Crown-5 and 12-crown-4 ethers complex Na^+ and Li^+, respectively. Make models of these crown ethers, and compare the sizes of the cavities.

18.7 Thiols and Sulfides

Thiols

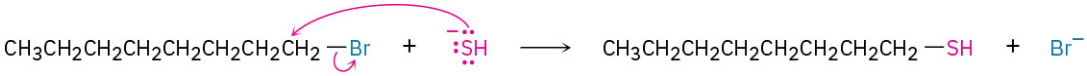
Thiols, also called **mercaptans**, are sulfur analogs of alcohols. They are named by the same system used for alcohols, with the suffix *-thiol* used in place of *-ol*. The –SH group itself is referred to as a **mercapto group**. Like alcohols, thiols are weakly acidic; the pK_a of CH_3SH, for instance, is 10.3. Unlike alcohols, however, thiols don't typically form hydrogen bonds because the sulfur atom is not sufficiently electronegative.



Ethanethiol **Cyclohexanethiol** ***m*-Mercaptobenzoic acid**

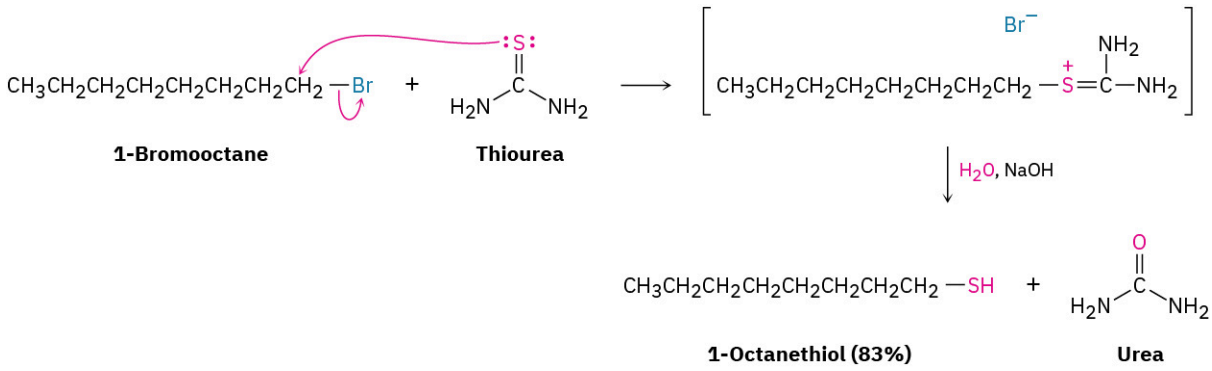
The most striking characteristic of thiols is their appalling odor. Skunk scent, for instance, is caused primarily by the simple thiols 3-methyl-1-butanethiol and 2-butene-1-thiol. Volatile thiols such as ethanethiol are also added to natural gas and liquefied propane to serve as an easily detectable warning in case of leaks.

Thiols are usually prepared from alkyl halides by S_N2 displacement with a sulfur nucleophile such as hydrosulfide anion, ^-SH.

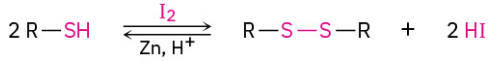


1-Bromooctane **1-Octanethiol (83%)**

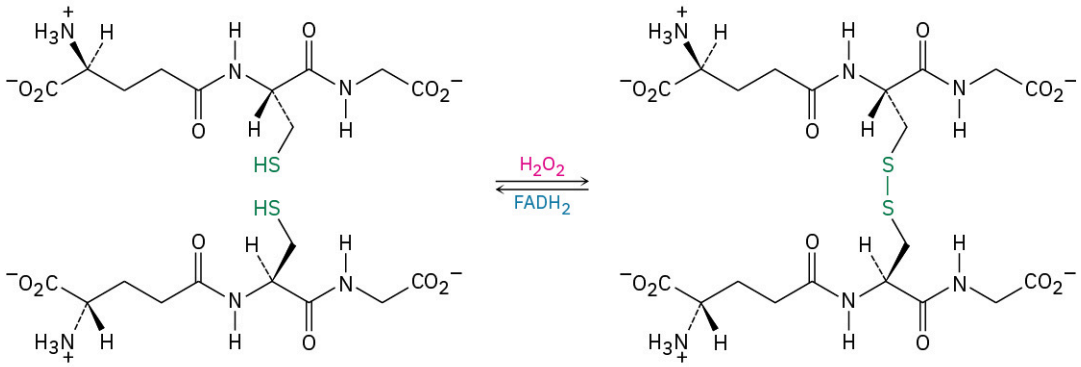
The reaction often works poorly unless an excess of the nucleophile is used because the product thiol can undergo a second S_N2 reaction with alkyl halide to give a **sulfide (R–S–R′)** as by-product. To circumvent this problem, thiourea, $(NH_2)_2C{=}S$, is often used as the nucleophile in the preparation of a thiol from an alkyl halide. The reaction occurs by displacement of the halide ion to yield an intermediate alkyl isothiourea salt, which is hydrolyzed by subsequent reaction with aqueous base.



Thiols can be oxidized by Br_2 or I_2 to yield **disulfides (R–S–S–R′)**. The reaction is easily reversed, and a disulfide can be reduced back to a thiol by treatment with zinc and acid.

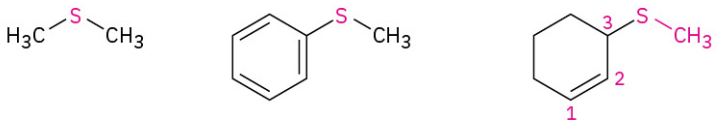


This thiol–disulfide interconversion is a key part of numerous biological processes. We'll see in Chapter 26, for instance, that disulfide formation is involved in defining the structure and three-dimensional conformations of proteins, where disulfide "bridges" often form cross-links between cysteine amino-acid units in the protein chains. Disulfide formation is also involved in the process by which cells protect themselves from oxidative degradation. A cellular component called *glutathione* removes potentially harmful oxidants and is itself oxidized to glutathione disulfide in the process. Reduction back to the thiol requires the coenzyme reduced flavin adenine dinucleotide, abbreviated $FADH_2$.

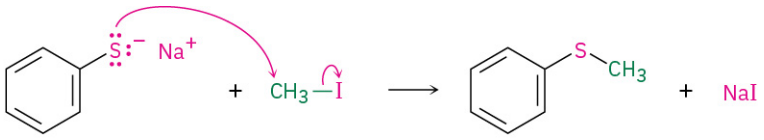


Sulfides

Sulfides are the sulfur analogs of ethers just as thiols are the sulfur analogs of alcohols. Sulfides are named by following the same rules used for ethers, with *sulfide* used in place of *ether* for simple compounds and *alkylthio* used in place of *alkoxy* for more complex substances.

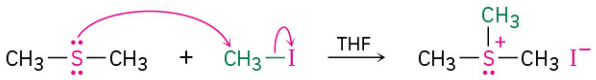


Treatment of a thiol with a base, such as NaH, gives the corresponding **thiolate ion** (RS^-), which undergoes reaction with a primary or secondary alkyl halide to give a sulfide. The reaction occurs by an S_N2 mechanism, analogous to the Williamson synthesis of ethers (**Section 18.2**).



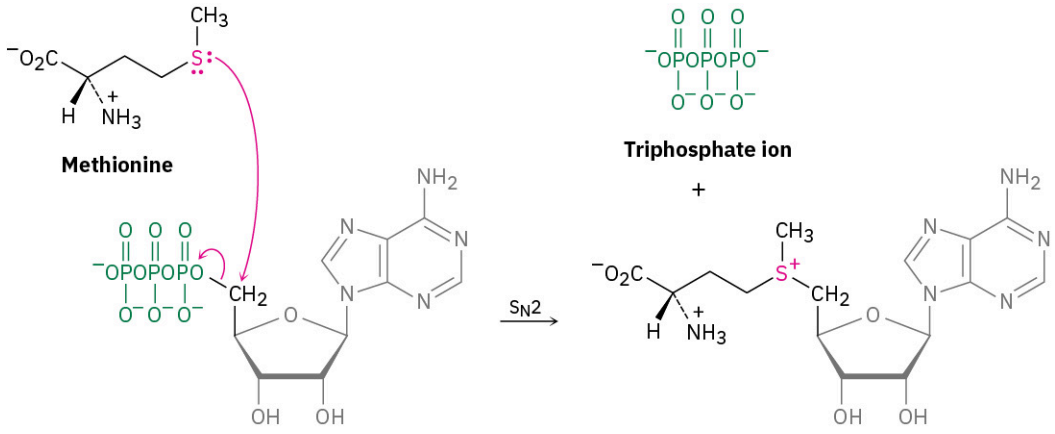
Sodium benzenethiolate **Methyl phenyl sulfide (96%)**

Despite their close structural similarity, sulfides and ethers differ substantially in their chemistry. Because the valence electrons on sulfur are farther from the nucleus and are less tightly held than those on oxygen (3*p* electrons versus 2*p* electrons), sulfur compounds are more nucleophilic than their oxygen analogs. Unlike dialkyl ethers, dialkyl sulfides react rapidly with primary alkyl halides by an S_N2 mechanism to give **sulfonium ions** (R_3S^+).



Dimethyl sulfide **Iodomethane** **Trimethylsulfonium iodide**

The most common example of this process in living organisms is the reaction of the amino acid methionine with adenosine triphosphate (ATP; Section 6.8) to give *S*-adenosylmethionine. The reaction is somewhat unusual in that the biological leaving group in this S_N2 process is the *triphosphate* ion rather than the more frequently seen diphosphate ion (Section 11.6).

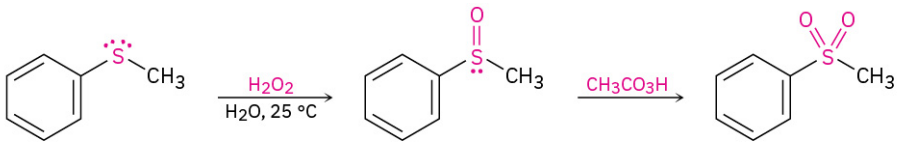


Methionine **Triphosphate ion**

Adenosine triphosphate (ATP) ***S*-Adenosylmethionine**

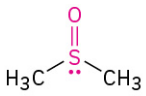
Sulfonium ions are themselves useful alkylating agents because a nucleophile can attack one of the groups bonded to the positively charged sulfur, displacing a neutral sulfide as leaving group. We saw an example of this in Section 11.6 (FIGURE 11.17), in which *S*-adenosylmethionine transferred a methyl group to norepinephrine to give adrenaline.

Another difference between sulfides and ethers is that sulfides are easily oxidized. Treatment of a sulfide with hydrogen peroxide, H_2O_2, at room temperature yields the corresponding **sulfoxide (R_2SO)**, and further oxidation of the sulfoxide with a peroxyacid yields a **sulfone (R_2SO_2)**.



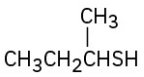
Methyl phenyl sulfide **Methyl phenyl sulfoxide** **Methyl phenyl sulfone**

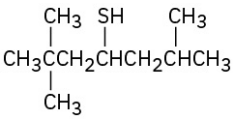
Dimethyl sulfoxide (DMSO) is a particularly well-known sulfoxide that is often used as a polar aprotic solvent. It must be handled with care, however, because it has a remarkable ability to penetrate the skin, carrying along whatever is dissolved in it.

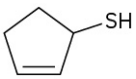


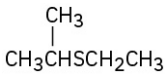
Dimethyl sulfoxide
(a polar aprotic solvent)

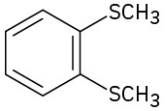
PROBLEM 18-16 Name the following compounds:

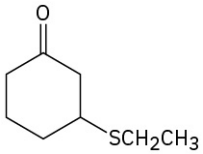
(a) 

(b) 

(c) 

(d) 

(e) 

(f) 

PROBLEM 18-17 2-Butene-1-thiol is one component of skunk spray. How would you synthesize this substance (either cis or trans) from methyl 2-butenoate? From 1,3-butadiene?



Methyl 2-butenoate **2-Butene-1-thiol**

18.8 Spectroscopy of Ethers

Infrared Spectroscopy

Ethers are difficult to identify by IR spectroscopy. Although they show an absorption due to C–O single-bond stretching in the range 1050 to 1150 cm^{-1}, many other kinds of absorptions occur in the same range. **FIGURE 18.3** shows the IR spectrum of diethyl ether and identifies the C–O stretch. Phenyl alkyl ethers show two strong absorbances for C–O stretching at 1050 and 1250 cm^{-1}. **FIGURE 18.4** shows the IR spectrum of anisole.

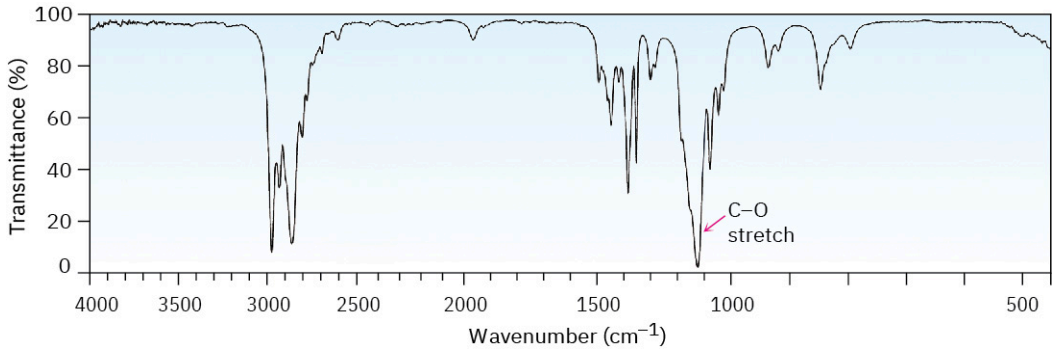


FIGURE 18.3 The infrared spectrum of diethyl ether, $CH_3CH_2OCH_2CH_3$.

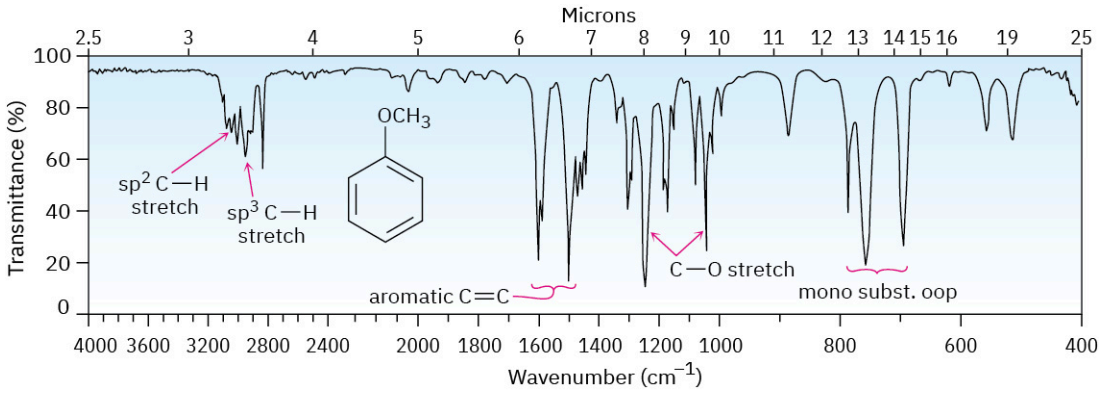


FIGURE 18.4 The infrared spectrum of anisole.

Nuclear Magnetic Resonance Spectroscopy

Hydrogens on carbon next to an ether oxygen are shifted downfield from the normal alkane resonance and show ^{1}H NMR absorptions in the region 3.4 to 4.5 δ. This downfield shift is clearly seen in the spectrum of dipropyl ether shown in **FIGURE 18.5**.

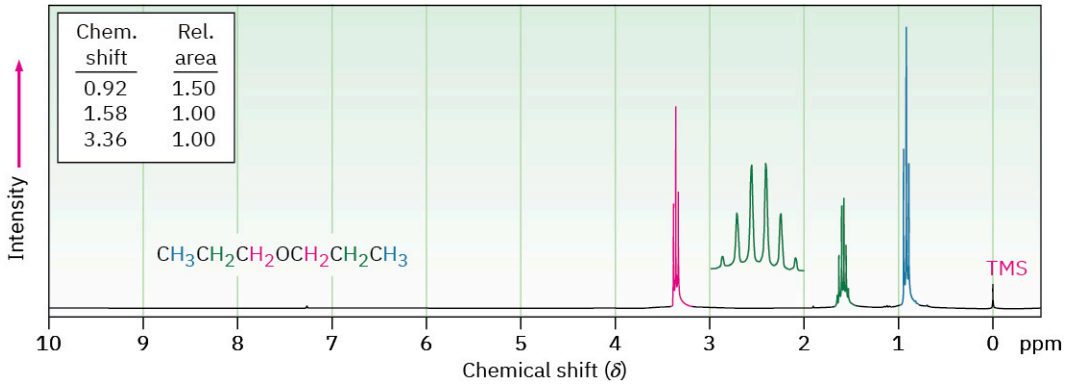


FIGURE 18.5 The ^{1}H NMR spectrum of dipropyl ether. Protons on carbon next to oxygen are shifted downfield to 3.4 δ.

Epoxides absorb at a slightly higher field than other ethers and show characteristic resonances at 2.5 to 3.5 δ in their ^{1}H NMR spectra, as indicated for 1,2-epoxypropane in **FIGURE 18.6**. The methylene protons of this epoxide are diastereotopic, and display complex splitting (see **Section 13.7** and **Section 13.8**).

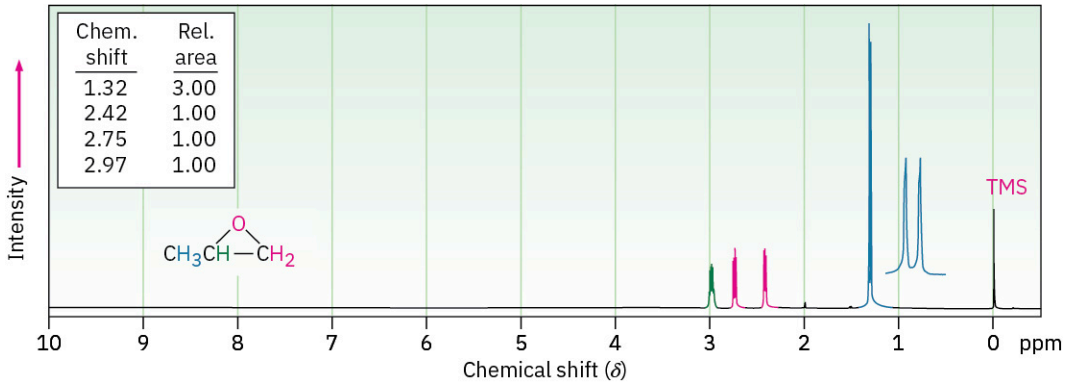
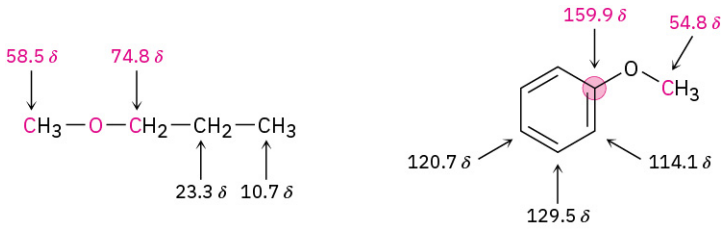
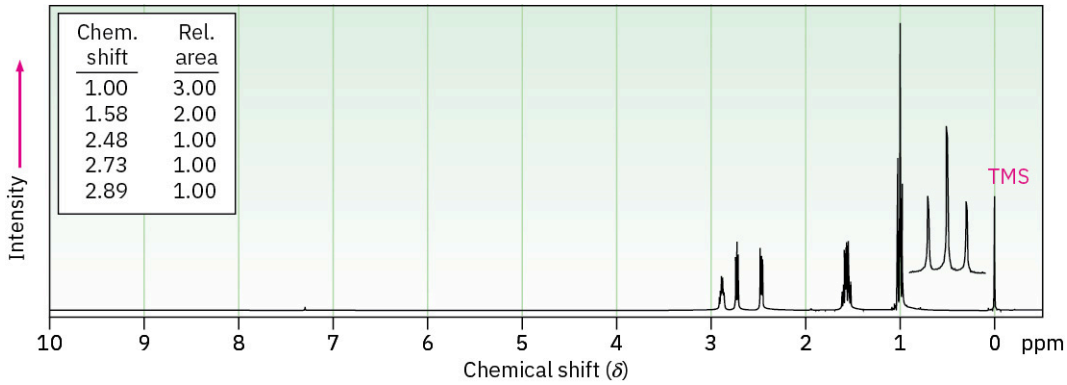


FIGURE 18.6 The ^{1}H NMR spectrum of 1,2-epoxypropane.

Ether carbon atoms also exhibit a downfield shift in the ^{13}C NMR spectrum, where they usually absorb in the 50 to 80 δ range. For example, the carbon atoms next to oxygen in methyl propyl ether absorb at 58.5 and 74.8 δ. Similarly, the methyl carbon in anisole absorbs at 54.8 δ.



PROBLEM 18-18 The ^{1}H NMR spectrum shown is that of a cyclic ether with the formula C_4H_8O. Propose a structure.


Chem. shift
Rel. area
1.00 3.00
1.58 2.00
2.48 1.00
2.73 1.00
2.89 1.00
Intensity
TMS
10 9 8 7 6 5 4 3 2 1 0 ppm
Chemical shift (δ)

CHEMISTRY MATTERS

Epoxy Resins and Adhesives

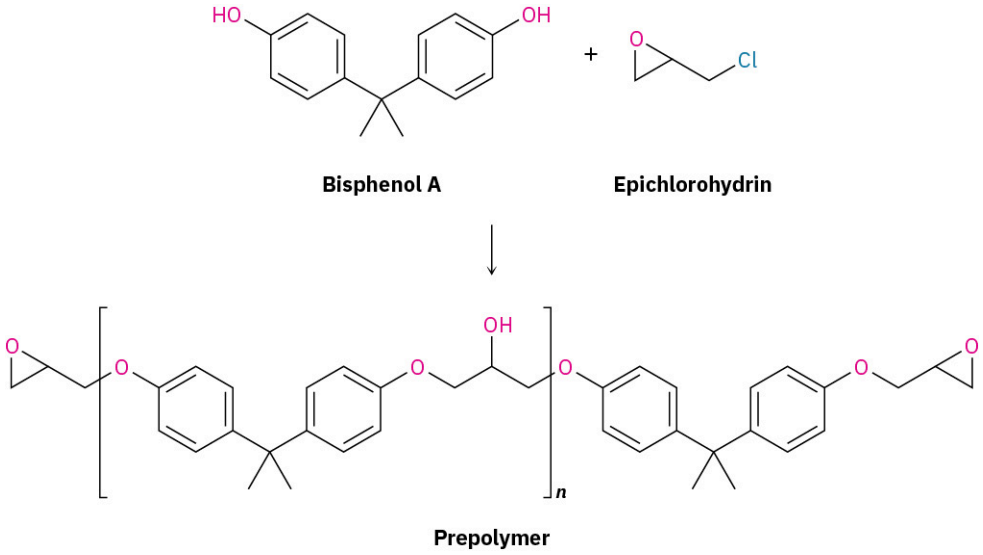
Few nonchemists know exactly what an epoxide is, but practically everyone has used an "epoxy glue" for household repairs or an epoxy resin for a protective coating. Worldwide, approximately 14 billion dollars' worth of epoxy resins and adhesives are used annually for a vast number of applications, including many in the aerospace industry. Much of the Boeing 787 Dreamliner, for instance, is held together with epoxy-based adhesives.

Epoxy resins and adhesives generally consist of two components that are mixed just prior to use. One component is a liquid "prepolymer," and the second is a "curing agent" that reacts with the prepolymer and causes it to solidify.

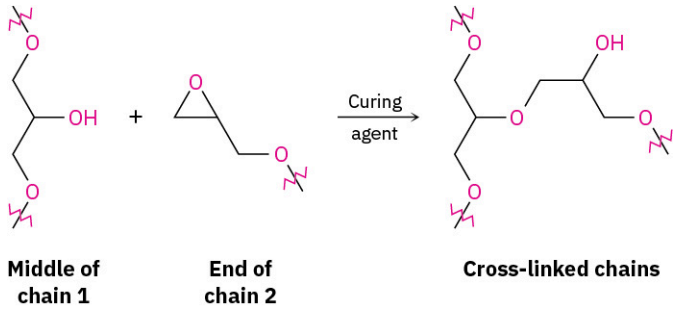
The most widely used epoxy resins and adhesives are based on a prepolymer made from bisphenol A and epichlorohydrin. On treatment with base, bisphenol A is converted into its anion, which acts as a nucleophile in an S_N2 reaction with epichlorohydrin. Each epichlorohydrin molecule can react with two molecules of bisphenol A, once by S_N2 displacement of chloride ion and once by nucleophilic opening of the epoxide ring. At the same time, each bisphenol A molecule can react with two epichlorohydrins, leading to a long polymer chain. Each end of a prepolymer chain has an unreacted epoxy group, and each chain has numerous secondary alcohol groups spaced regularly along its midsection.



FIGURE 18.7 Kayaks are often made of a high-strength polymer coated with epoxy resin. (credit: "Potomac River Festival Virginia 2017 Great Falls Kayak Race" by Watts/Flickr, CC BY 2.0)



When an epoxide is to be used, a basic curing agent such as a tertiary amine, R_3N, is added to cause the individual prepolymer chains to link together. This cross-linking of chains is simply a base-catalyzed, S_N2 epoxide ring-opening of an –OH group in the middle of one chain with an epoxide group on the end of another chain. The result of such cross-linking is formation of a vast, three-dimensional tangle that has enormous strength and chemical resistance.



Key Terms

- **alkoxymercuration**
- **crown ether**
- **disulfide (R–S–S–R′)**
- **ether (R–O–R′)**
- **mercapto group**
- **sulfide (R–S–R′)**
- **sulfone**
- **sulfonium ion**
- **sulfoxide**
- **thiol**
- **thiolate ion (RS^-)**
- **Williamson ether synthesis**

Summary

This chapter has finished the coverage of functional groups with C–O and C–S single bonds, focusing primarily on ethers, epoxides, thiols, and sulfides. **Ethers** are compounds that have two organic groups bonded to the same oxygen atom, ROR′. The organic groups can be alkyl, vinylic, or aryl, and the oxygen atom can be in a ring or in an open chain. Ethers are prepared by either Williamson ether synthesis, which involves S_N2 reaction of an alkoxide ion with a primary alkyl halide, or the **alkoxymercuration** reaction, which involves Markovnikov addition of an alcohol to an alkene.

Ethers are inert to most reagents but react with strong acids to give cleavage products. Both HI and HBr are often used. The cleavage reaction takes place by an S_N2 mechanism at the less highly substituted site if only primary and secondary alkyl groups are bonded to the ether oxygen, but by an S_N1 or E1 mechanism if one of the alkyl groups bonded to oxygen is tertiary.

Epoxides are cyclic ethers with a three-membered, oxygen-containing ring. Because of the strain in the ring, epoxides undergo a cleavage reaction with both acids and bases. Acid-catalyzed ring-opening occurs with a regiochemistry that depends on the structure of the epoxide. Cleavage of the C–O bond at the less highly substituted site occurs if both epoxide carbons are primary or secondary, but cleavage of the C–O bond to the more highly substituted site occurs if one of the epoxide carbons is tertiary. Base-catalyzed epoxide ring-opening occurs by S_N2 reaction of a nucleophile at the less hindered epoxide carbon.

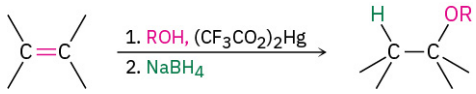
Thiols, the sulfur analogs of alcohols, are usually prepared by S_N2 reaction of an alkyl halide with thiourea. Mild oxidation of a thiol yields a **disulfide**, and mild reduction of a disulfide returns the thiol. **Sulfides**, the sulfur analogs of ethers, are prepared by an S_N2 reaction between a thiolate anion and a primary or secondary alkyl halide. Sulfides are more nucleophilic than ethers and can be alkylated by reaction with a primary alkyl halide to yield a **sulfonium ion**. Sulfides can also be oxidized to **sulfoxides** and to **sulfones**.

Summary of Reactions

1. Synthesis of ethers (Section 18.2)
   a. Williamson ether synthesis

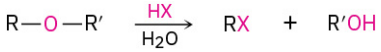


b. Alkoxymercuration/demercuration

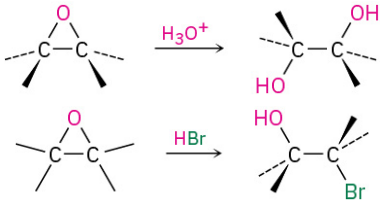


2. Reactions of ethers

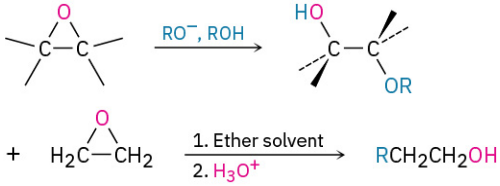
a. Cleavage by HBr or HI (**Section 18.3**)



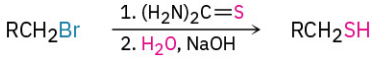
b. Acid-catalyzed epoxide opening (**Section 18.5**)



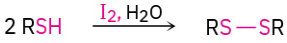
c. Base-catalyzed epoxide opening (**Section 18.5**)



3. Synthesis of thiols (**Section 18.7**)



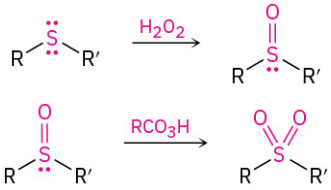
4. Oxidation of thiols to disulfides (**Section 18.7**)



5. Synthesis of sulfides (**Section 18.7**)



6. Oxidation of sulfides to sulfoxides and sulfones (**Section 18.7**)

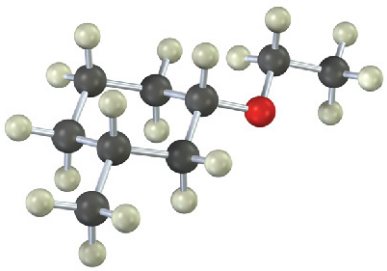


Additional Problems

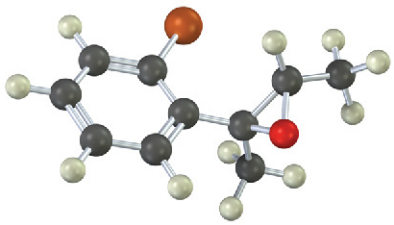
Visualizing Chemistry

PROBLEM 18-19 Give IUPAC names for the following compounds (red = O; reddish brown = Br; yellow = S):

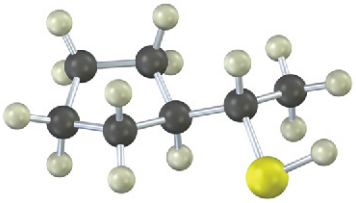
(a)



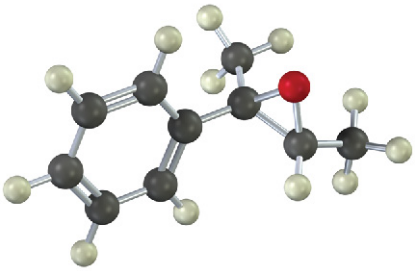
(b)



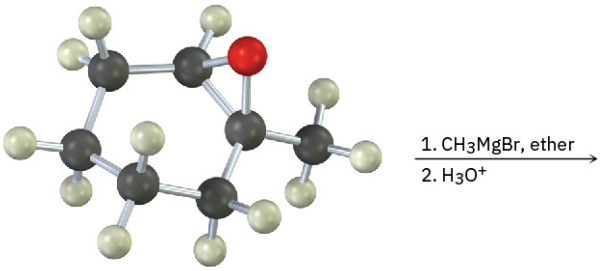
(c)



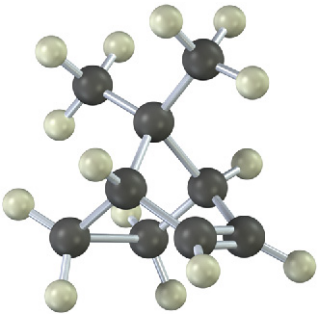
PROBLEM 18-20 Show the product, including stereochemistry, that would result from reaction of the following epoxide with HBr:



PROBLEM 18-21 Show the product, including stereochemistry, of the following reaction:



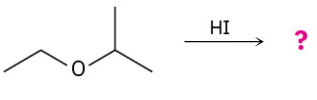
PROBLEM 18-22 Treatment of the following alkene with a peroxyacid yields an epoxide different from that obtained by reaction with aqueous Br_2 followed by base treatment. Propose structures for the two epoxides, and explain the result.



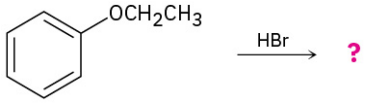
Mechanism Problems

PROBLEM 18-23 Predict the product(s) and provide the mechanism for each of the following reactions.

(a)

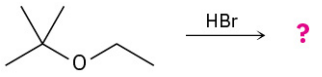


(b)

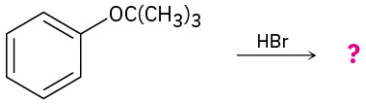


PROBLEM 18-24 Predict the product(s) and provide the mechanism for each of the following reactions.

(a)

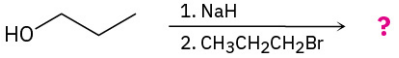


(b)

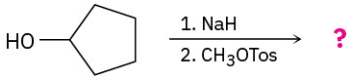


PROBLEM 18-25 Predict the product(s) and provide the mechanism for each of the following two-step processes.

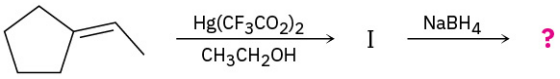
(a)



(b)

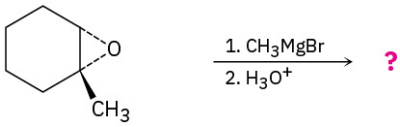


PROBLEM 18-26 The alkoxymercuration of alkenes involves the formation of an organomercury intermediate (I), which is reduced with $NaBH_4$ to give an ether product. Predict the ether product and provide the mechanism for the following reaction.

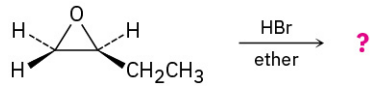


PROBLEM 18-27 Predict the product(s) and provide the mechanism for the following reactions:

(a)

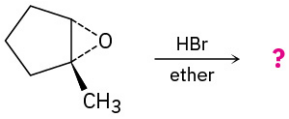


(b)

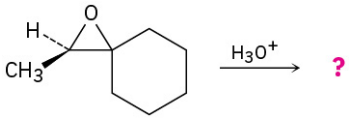


PROBLEM 18-28 Predict the product(s) and provide the mechanism for each of the following reactions.

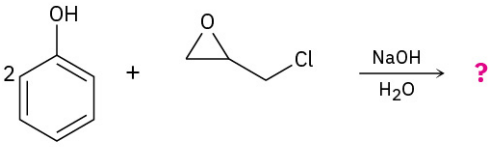
(a)



(b)

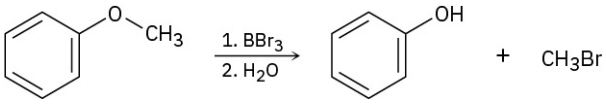


PROBLEM 18-29 In the formation of the prepolymer used to make epoxy resins, a bisphenol reacts with epichlorohydrin in the presence of a base. Show the product and mechanism when two moles of phenol react with epichlorohydrin.

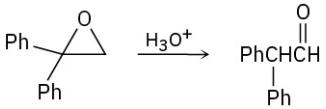


Epichlorohydrin

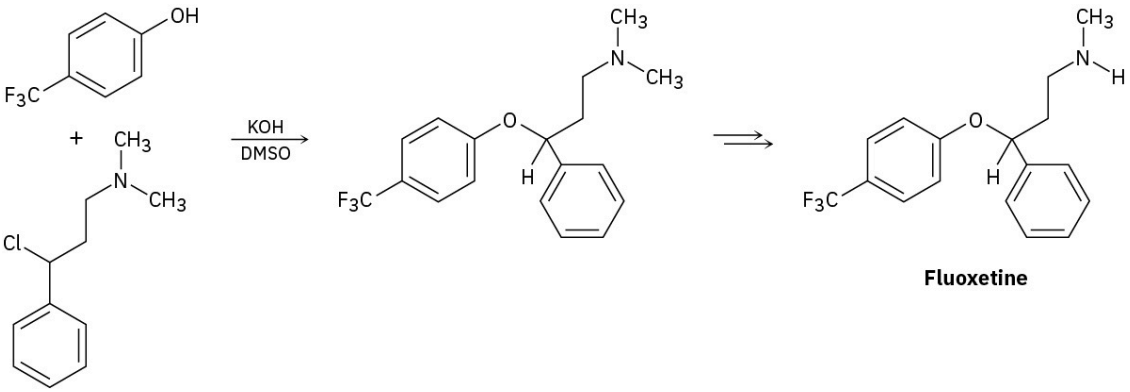
PROBLEM 18-30 Ethers undergo an acid-catalyzed cleavage reaction when treated with the Lewis acid BBr_3 at room temperature. Propose a mechanism for the reaction.



PROBLEM 18-31 Treatment of 1,1-diphenyl-1,2-epoxyethane with aqueous acid yields diphenyl acetaldehyde as the major product. Propose a mechanism for the reaction.



PROBLEM 18-32 Fluoxetine, a heavily prescribed antidepressant marketed under the name Prozac, can be prepared by a route that begins with reaction between a phenol and an alkyl chloride.

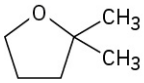


Fluoxetine

(a) The rate of the reaction depends on both phenol and alkyl halide. Is this an S_N1 or an S_N2 reaction? Show the mechanism.

(b) The physiologically active enantiomer of fluoxetine has (*S*) stereochemistry. Based on your answer in part **(a)**, draw the structure of the alkyl chloride you would need, showing the correct stereochemistry.

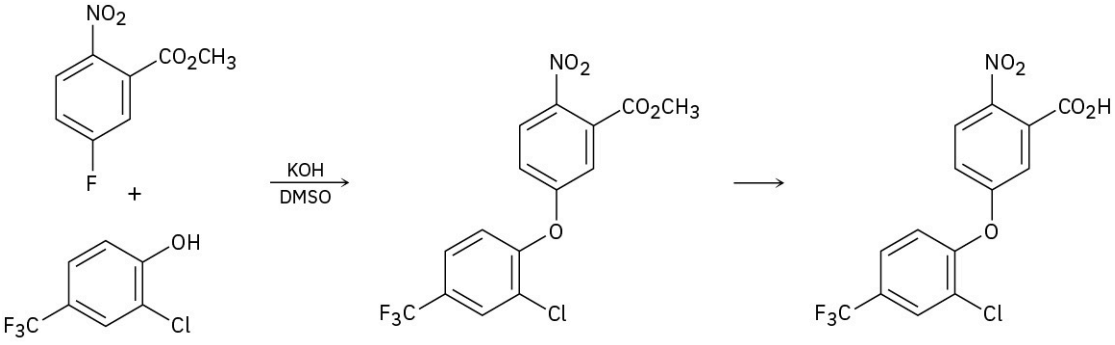
PROBLEM 18-33 When 2-methyl-2,5-pentanediol is treated with sulfuric acid, dehydration occurs and 2,2-dimethyltetrahydrofuran is formed. Suggest a mechanism for this reaction. Which of the two oxygen atoms is most likely to be eliminated, and why?



2,2-Dimethyltetrahydrofuran

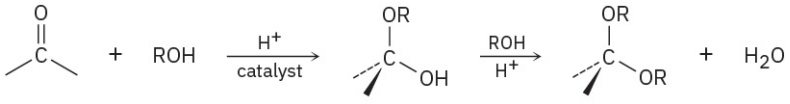
PROBLEM 18-34 Methyl aryl ethers, such as anisole, are cleaved to iodomethane and a phenoxide ion by treatment with LiI in hot DMF. Propose a mechanism for this reaction.

PROBLEM 18-35 The herbicide acifluorfen can be prepared by a route that begins with reaction between a phenol and an aryl fluoride. Propose a mechanism.



Acifluorfen

PROBLEM 18-36 Aldehydes and ketones undergo acid-catalyzed reaction with alcohols to yield *hemiacetals*, from aldehydes or ketals with ketones compounds that have one alcohol-like oxygen and one ether-like oxygen bonded to the same carbon. Further reaction of a hemiacetal with alcohol then yields an *acetal,* a compound that has two ether-like oxygens bonded to the same carbon.

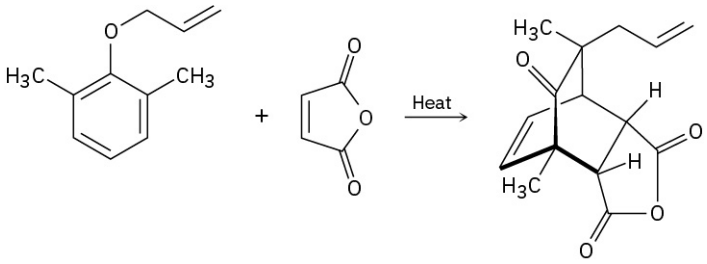


A ketone **A hemiketal** **A ketal**

(a) Show the structures of the hemiketal and ketal you would obtain by reaction of cyclohexanone with ethanol.

(b) Propose a mechanism for the conversion of a hemiacetal into a ketal.

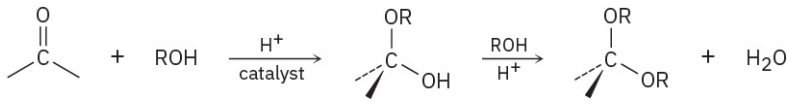
PROBLEM 18-37 Propose a mechanism to account for the following transformation. What two kinds of reactions are occurring?



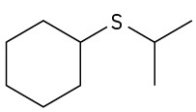
Naming Ethers

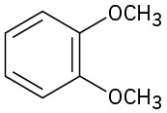
PROBLEM 18-38 Draw structures corresponding to the following IUPAC names:
(a) Ethyl 1-ethylpropyl ether **(b)** Di(*p*-chlorophenyl) ether **(c)** 3,4-Dimethoxybenzoic acid
(d) Cyclopentyloxycyclohexane **(e)** 4-Allyl-2-methoxyphenol (eugenol; from oil of cloves)

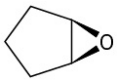
PROBLEM 18-39 Give IUPAC names for the following structures:

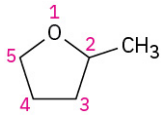


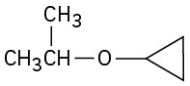
A ketone **A hemiketal** **A ketal**

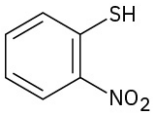
(a)


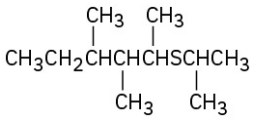
(b)


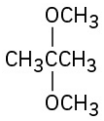
(c)


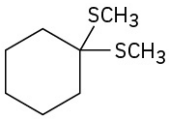
(d)


(e)


(f)


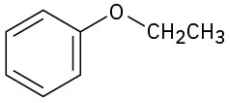
(g)


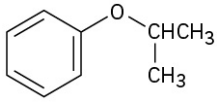
(h)


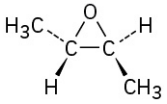
(i)


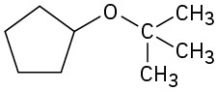
Synthesizing Ethers

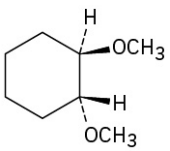
PROBLEM 18-40 How would you prepare the following ethers?

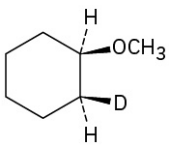
(a)


(b)


(c)


(d)


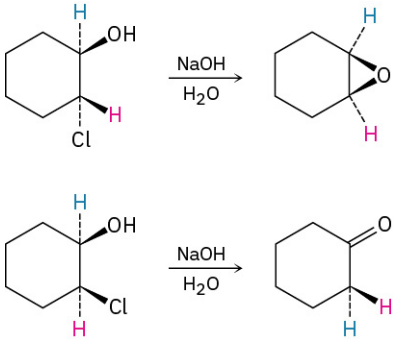
(e)


(f)


PROBLEM 18-41 How would you prepare the following compounds from 1-phenylethanol?
(a) Methyl 1-phenylethyl ether **(b)** Phenylepoxyethane **(c)** *tert*-Butyl 1-phenylethyl ether
(d) 1-Phenylethanethiol

PROBLEM 18-42 *tert*-Butyl ethers can be prepared by the reaction of an alcohol with 2-methylpropene in the presence of an acid catalyst. Propose a mechanism for this reaction.

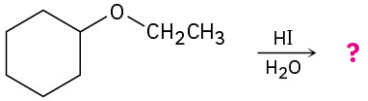
PROBLEM 18-43 Treatment of *trans*-2-chlorocyclohexanol with NaOH yields 1,2-epoxycyclohexane, but reaction of the cis isomer under the same conditions yields cyclohexanone. Propose mechanisms for both reactions, and explain why the different results are obtained.



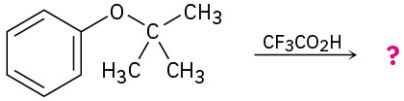
Reactions of Ethers and Epoxides

PROBLEM 18-44 Predict the products of the following ether cleavage reactions:

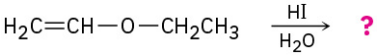
(a)



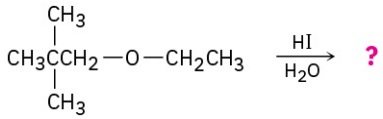
(b)



(c)

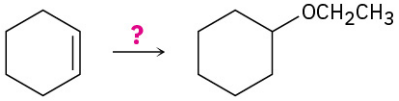


(d)

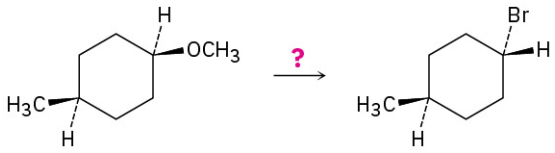


PROBLEM 18-45 How would you carry out the following transformations? More than one step may be required.

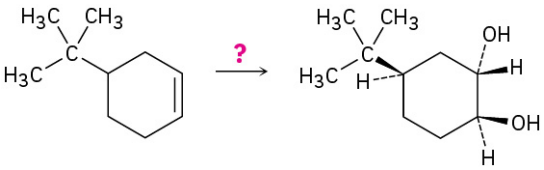
(a)



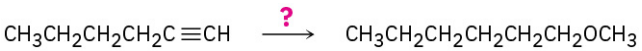
(b)



(c)



(d)



(e)



PROBLEM 18-46 What product would you expect from cleavage of tetrahydrofuran with HI?

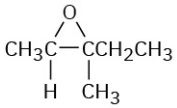
PROBLEM 18-47 Write the mechanism of the hydrolysis of *cis*-5,6-epoxydecane by reaction with aqueous acid. What is the stereochemistry of the product, assuming normal backside S_N2 attack?

PROBLEM 18-48 What is the stereochemistry of the product from acid-catalyzed hydrolysis of *trans*-5,6-epoxydecane? How does the product differ from that formed in Problem 18-47?

PROBLEM 18-49 Acid-catalyzed hydrolysis of a 1,2-epoxycyclohexane produces a trans-diaxial 1,2-diol. What product would you expect to obtain from acidic hydrolysis of *cis*-3-*tert*-butyl-1,2-epoxycyclohexane? (Recall that the bulky *tert*-butyl group locks the

cyclohexane ring into a specific conformation.)

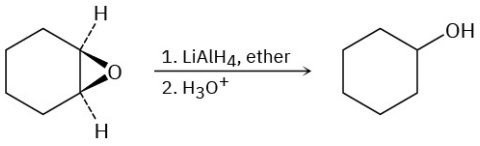
PROBLEM 18-50 Imagine that you have treated (2*R*,3*R*)-2,3-epoxy-3-methylpentane with aqueous acid to carry out a ring-opening reaction.



2,3-Epoxy-3-methylpentane (no stereochemistry implied)

(a) Draw the epoxide, showing stereochemistry.
(b) Draw and name the product, showing stereochemistry. **(c)** Is the product chiral? Explain.
(d) Is the product optically active? Explain.

PROBLEM 18-51 Epoxides are reduced by treatment with lithium aluminum hydride to yield alcohols. Propose a mechanism for this reaction.



PROBLEM 18-52 Show the structure and stereochemistry of the alcohol that would result if 1,2-epoxycyclohexane were reduced with lithium aluminum deuteride, $LiAlD_4$ (Problem 18-51).

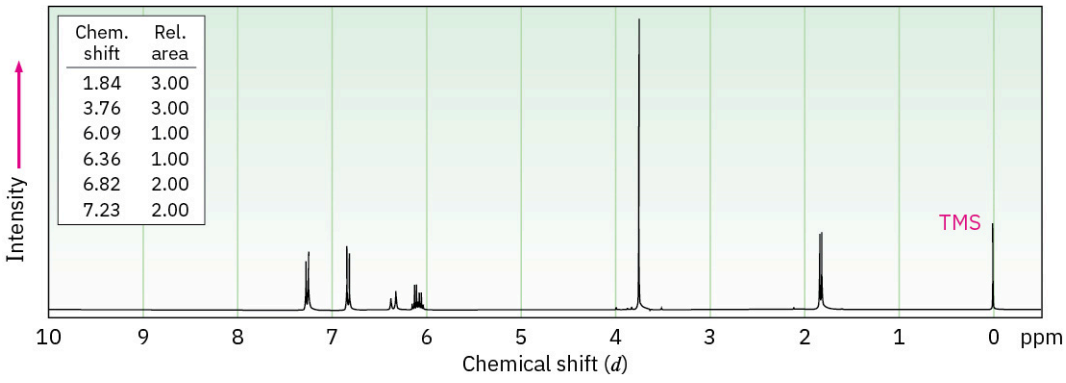
Spectroscopy

PROBLEM 18-53 The red fox *(Vulpes vulpes)* uses a chemical communication system based on scent marks in urine. One component of fox urine is a sulfide whose mass spectrum has $M^+ = 116$. IR spectroscopy shows an intense band at 890 cm^{-1}, and ^1H NMR spectroscopy reveals the following peaks:

1.74 δ (3 H, singlet); 2.11 δ (3 H, singlet); 2.27 δ (2 H, triplet, J = 4.2 Hz); 2.57 δ (2 H, triplet, J = 4.2 Hz); 4.73 δ (2 H, broad)

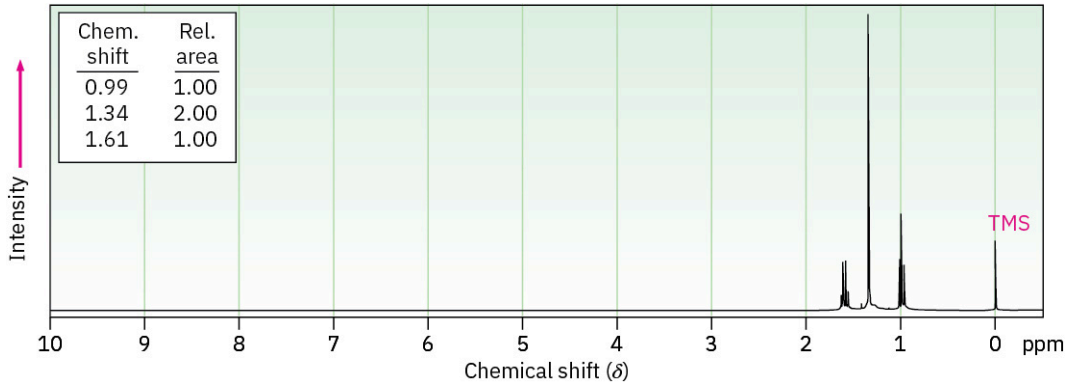
Propose a structure consistent with these data. [Note: $(CH_3)_2S$ absorbs at 2.1 δ].

PROBLEM 18-54 Anethole, $C_{10}H_{12}O$, a major constituent of the oil of anise, has the ^1H NMR spectrum shown. On careful oxidation with $Na_2Cr_2O_7$, anethole yields *p*-methoxybenzoic acid. What is the structure of anethole? Assign all peaks in the NMR spectrum, and account for the observed splitting patterns.

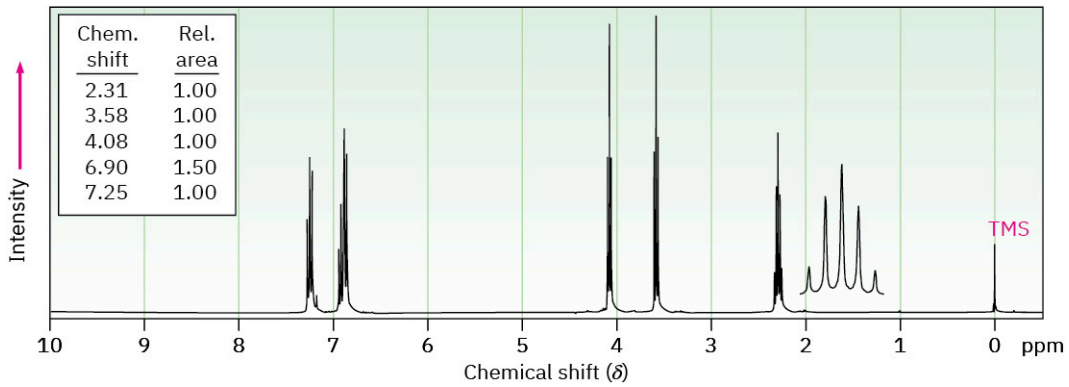


PROBLEM 18-55 Propose structures for compounds that have the following ^1H NMR spectra:

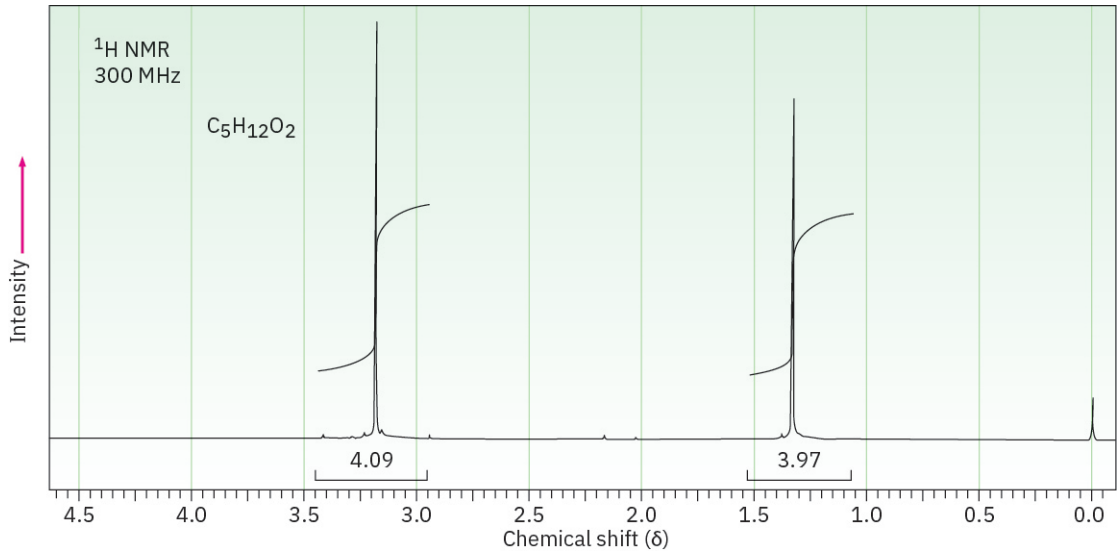
(a) $C_5H_{12}S$ (An –SH proton absorbs near 1.6 δ.)



(b) $C_9H_{11}BrO$



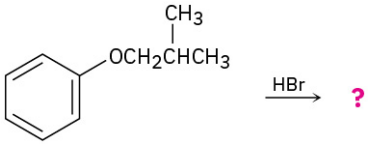
(c) $C_5H_{12}O_2$



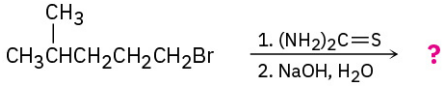
General Problems

PROBLEM 18-56 Predict the products of the following reactions:

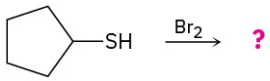
(a)



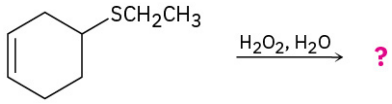
(b)



(c)



(d)



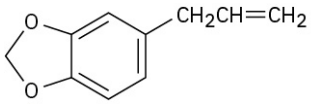
PROBLEM 18-57 How would you synthesize anethole (Problem 18-54) from phenol?

PROBLEM 18-58 How could you prepare benzyl phenyl ether from benzene and phenol? More than one step is required.

PROBLEM 18-59 *Meerwein's reagent,* triethyloxonium tetrafluoroborate, is a powerful ethylating agent that converts alcohols into ethyl ethers at neutral pH. Show the reaction of Meerwein's reagent with cyclohexanol, and account for the fact that trialkyloxonium salts are much more reactive alkylating agents than alkyl iodides.

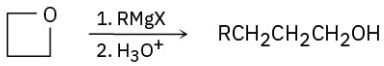
$(CH_3CH_2)_3O^+\ BF_4^-$ **Meerwein's reagent**

PROBLEM 18-60 Safrole, a substance isolated from oil of sassafras, is used as a perfumery agent. Propose a synthesis of safrole from catechol (1,2-benzenediol).



Safrole

PROBLEM 18-61 Grignard reagents react with oxetane, a four-membered cyclic ether, to yield primary alcohols, but the reaction is much slower than the corresponding reaction with ethylene oxide. Suggest a reason for the difference in reactivity between oxetane and ethylene oxide.



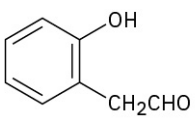
Oxetane

PROBLEM 18-62 The *Zeisel method* is an old analytical procedure for determining the number of methoxyl groups in a compound. A weighed amount of the compound is heated with concentrated HI, ether cleavage occurs, and the iodomethane product is distilled off and passed into an alcohol solution of $AgNO_3$, where it reacts to form a precipitate of silver iodide. The AgI is then collected and weighed, and the percentage of methoxyl groups in the sample is thereby determined. For example, 1.06 g of vanillin, the material responsible for the characteristic odor of vanilla, yields 1.60 g of AgI. If vanillin has a molecular weight of 152, how many methoxyl groups does it contain?

PROBLEM 18-63 Disparlure, $C_{19}H_{38}O$, is a sex attractant released by the female spongy moth, *Lymantria dispar.* The ^1H NMR spectrum of disparlure shows a large absorption in the alkane region, 1 to 2 δ, and a triplet at 2.8 δ. Treatment of disparlure, first with aqueous acid and then with $KMnO_4$, yields two carboxylic acids identified as undecanoic acid and 6-methylheptanoic acid. ($KMnO_4$ cleaves 1,2-diols to yield carboxylic acids.) Neglecting stereochemistry, propose a structure for disparlure. The actual compound is a chiral molecule with 7*R*,8*S* stereochemistry. Draw disparlure, showing the correct stereochemistry.

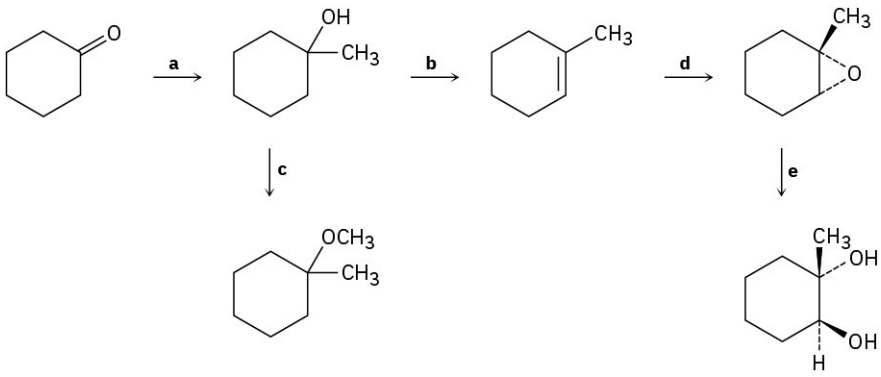
PROBLEM 18-64 How would you synthesize racemic disparlure (Problem 18-63) from compounds having ten or fewer carbons?

PROBLEM 18-65 How would you prepare *o*-hydroxyphenylacetaldehyde from phenol? More than one step is required.



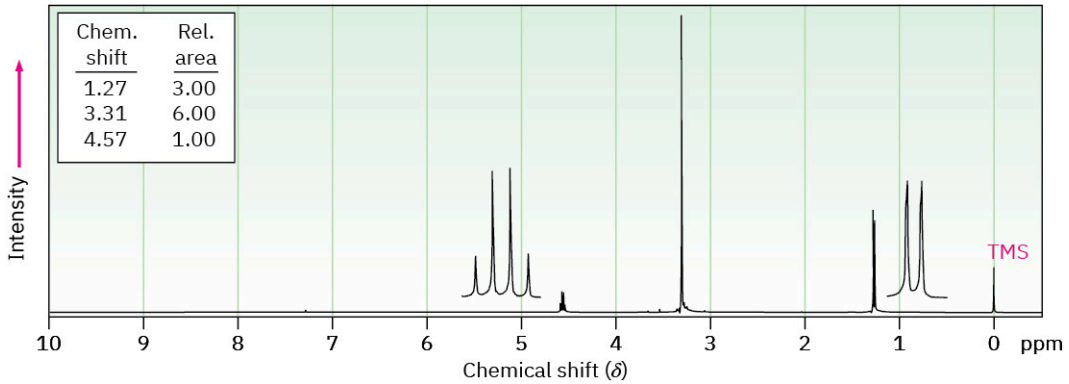
***o*-Hydroxyphenylacetaldehyde**

PROBLEM 18-66 Identify the reagents **a**–**e** in the following scheme:

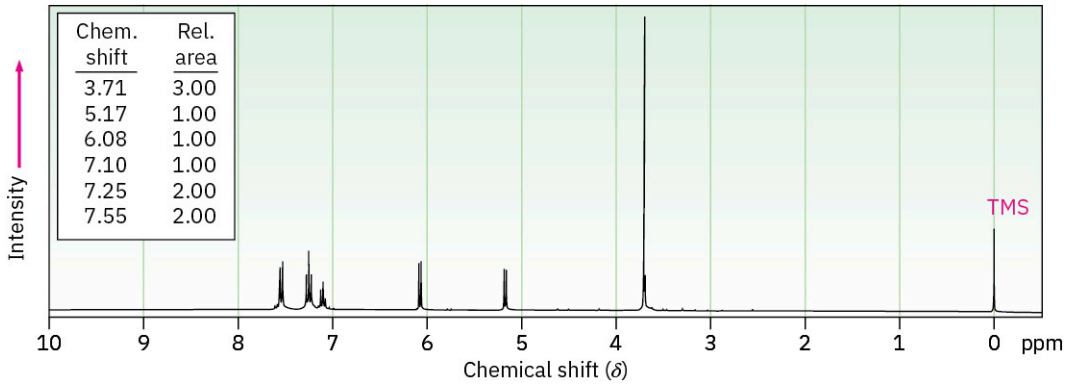


PROBLEM 18-67 Propose structures for compounds that have the following ^{1}H NMR spectra:

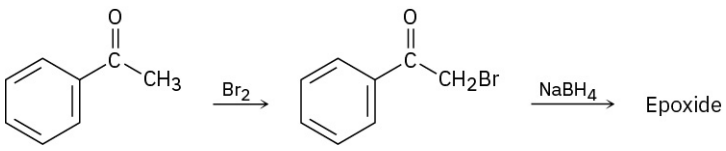
(a) $C_4H_{10}O_2$



(b) $C_9H_{10}O$



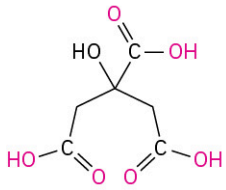
PROBLEM 18-68 We saw in **Section 17.4** that ketones react with $NaBH_4$ to yield alcohols. We'll also see in **Section 22.3** that ketones react with Br_2 to yield α-bromo ketones. Perhaps surprisingly, treatment with $NaBH_4$ of the α-bromo ketone from acetophenone yields an epoxide rather than a bromo alcohol. Show the structure of the epoxide, and explain its formation.



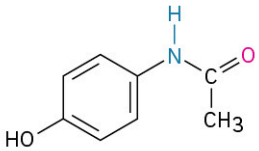
Acetophenone **An α-bromo ketone**

Preview of Carbonyl Chemistry

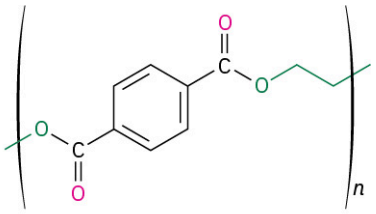
Carbonyl compounds are everywhere. Most biological molecules contain carbonyl groups, as do most pharmaceutical agents and many of the synthetic chemicals that affect our everyday lives. Citric acid, found in lemons and oranges; acetaminophen, the active ingredient in many over-the-counter headache remedies; and Dacron, the polyester material used in clothing, all contain different kinds of carbonyl groups.



Citric acid
(a carboxylic acid)



Acetaminophen
(an amide)



Dacron
(a polyester)

To a great extent, the chemistry of living organisms is the chemistry of carbonyl compounds. Thus, we'll spend the next five chapters discussing the chemistry of the **carbonyl group, C=O** (pronounced car-bo-**neel**). There are many different kinds of carbonyl compounds and many different reactions, but there are only a few fundamental principles that tie the entire field together. The purpose of this brief preview is not to show details of specific reactions but rather to provide a framework for learning carbonyl-group chemistry. Read through this preview now, and return to it on occasion to remind yourself of the larger picture.

I Kinds of Carbonyl Compounds

TABLE 18.2 shows some of the many different kinds of carbonyl compounds. All contain an **acyl group** (R–C=O) bonded to another substituent. The R part of the acyl group can be practically any organic part/structure, and the other substituent to which the acyl group is bonded might be a carbon, hydrogen, oxygen, halogen, nitrogen, or sulfur.

TABLE 18.2 Some Types of Carbonyl Compounds

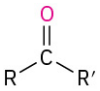
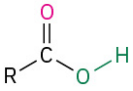
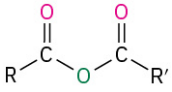
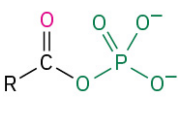
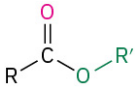
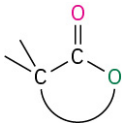
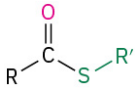
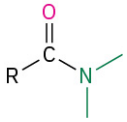
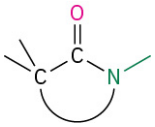
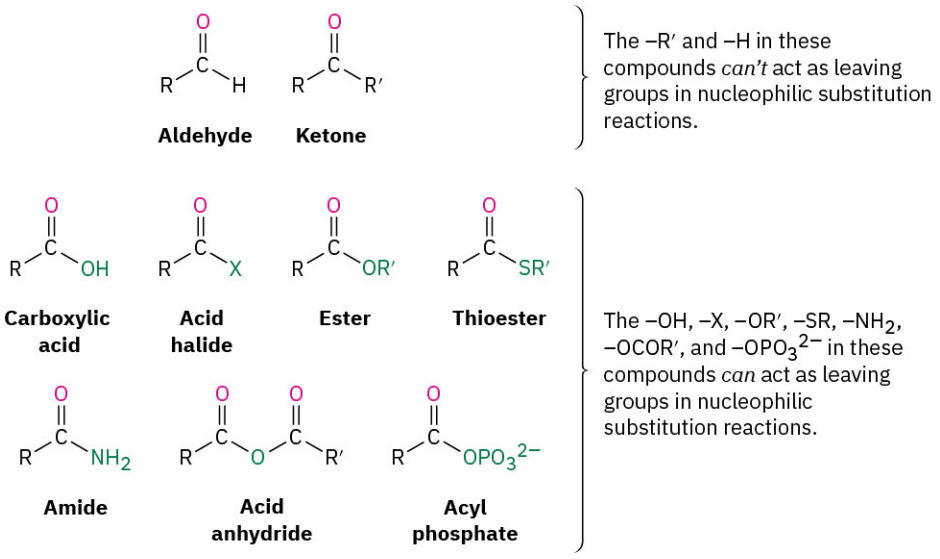
Name	General formula	Name ending
Aldehyde		*-al*
Ketone		*-one*
Carboxylic acid		*-oic acid*
Acid halide		*-yl or -oyl halide*
Acid anhydride		*-oic anhydride*
Acyl phosphate		*-yl phosphate*

TABLE 18.2 Some Types of Carbonyl Compounds

Name	General formula	Name ending
Ester		*-oate*
Lactone (cyclic ester)		None
Thioester		*-thioate*
Amide		*-amide*
Lactam (cyclic amide)		None

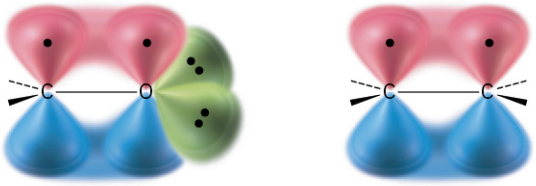
It's useful to classify carbonyl compounds into two categories based on the kinds of chemistry they undergo. In one category are aldehydes and ketones; in the other are carboxylic acids and their derivatives. The acyl group in an aldehyde or ketone is bonded to an atom (H or C, respectively) that can't stabilize a negative charge and therefore can't act as a leaving group in a nucleophilic substitution reaction. The acyl group in a carboxylic acid or its derivative, however, is bonded to an atom (oxygen, halogen, sulfur, nitrogen) that *can* stabilize a negative charge and therefore *can* act as a leaving group in a nucleophilic substitution reaction.



II Nature of the Carbonyl Group

The carbon–oxygen double bond of a carbonyl group is similar in many respects to the carbon–carbon double bond of an alkene. The carbonyl carbon atom is sp^2-hybridized and forms three σ bonds. The fourth valence electron remains in a carbon p orbital and forms a π bond to oxygen by overlapping with an oxygen p orbital.

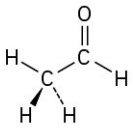
The oxygen atom also has two nonbonding pairs of electrons, which occupy its remaining two orbitals.



Carbonyl group **Alkene**

Like alkenes, carbonyl compounds are planar about the double bond and have bond angles of approximately 120°. FIGURE 18.8 shows the structure of acetaldehyde and indicates its bond lengths and angles. As you might expect, the carbon–oxygen double bond is both shorter (122 pm versus 143 pm) and stronger [732 kJ/mol (175 kcal/mol) versus 385 kJ/mol (92 kcal/mol)] than a C–O single bond.

Bond angle	(°)	Bond length	(pm)
H—C—C	118	C=O	122
C—C=O	121	C—C	150
H—C=O	121	OC—H	109



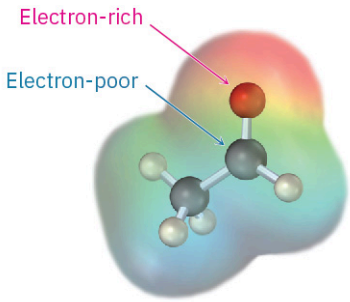


FIGURE 18.8 **Structure of acetaldehyde.**

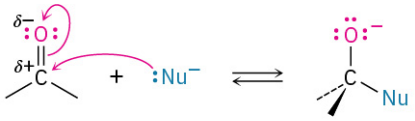
As indicated by the electrostatic potential map in FIGURE 18.8, the carbon–oxygen double bond is strongly polarized because of the high electronegativity of oxygen relative to carbon. Thus, the carbonyl carbon atom carries a partial positive charge, is an electrophilic (Lewis acidic) site, and reacts with nucleophiles. Conversely, the carbonyl oxygen atom carries a partial negative charge, is a nucleophilic (Lewis basic) site, and reacts with electrophiles. We'll see in the next five chapters that the majority of carbonyl-group reactions can be rationalized by simple polarity arguments.

III General Reactions of Carbonyl Compounds

Both in the laboratory and in living organisms, most reactions of carbonyl compounds take place by one of four general mechanisms: *nucleophilic addition, nucleophilic acyl substitution, alpha substitution,* and *carbonyl condensation.* These mechanisms have many variations, just as alkene electrophilic addition reactions and S_N2 reactions do, but the variations are much easier to learn when the fundamental features of the mechanisms are made clear. Let's see what the four mechanisms are and what kinds of chemistry carbonyl compounds undergo.

Nucleophilic Addition Reactions of Aldehydes and Ketones (Chapter 19)

The most common reaction of aldehydes and ketones is the **nucleophilic addition reaction**, in which a nucleophile, **:**Nu^-, adds to the electrophilic carbon of the carbonyl group. Because the nucleophile uses an electron pair to form a new bond to carbon, two electrons from the carbon–oxygen double bond must move toward the electronegative oxygen atom to give an alkoxide anion. The carbonyl carbon rehybridizes from sp^2 to sp^3 during the reaction, and the alkoxide ion product therefore has tetrahedral geometry.



A carbonyl compound (sp^2-hybridized carbon) **A tetrahedral intermediate (sp^3-hybridized carbon)**

Once formed, and depending on the nature of the nucleophile, the tetrahedral alkoxide intermediate can undergo one of two further reactions, as shown in FIGURE 18.9. Often, the tetrahedral alkoxide intermediate is simply protonated by water or acid to form an alcohol product. Alternatively, the tetrahedral intermediate

can be protonated and expel the oxygen to form a new double bond between the carbonyl carbon and the nucleophile. We'll study both processes in detail in Chapter 19.

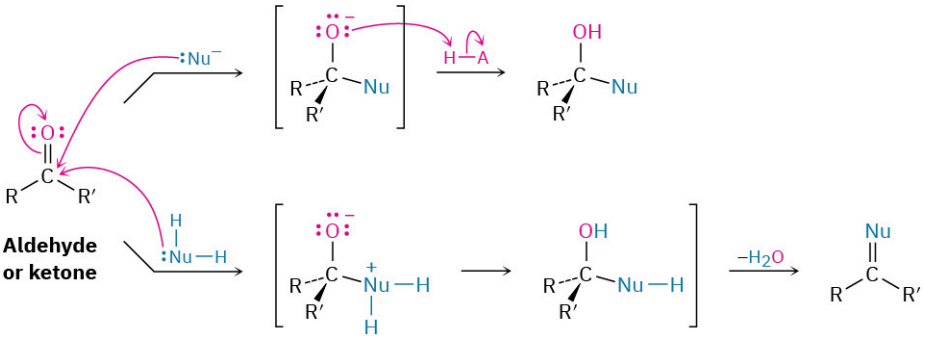
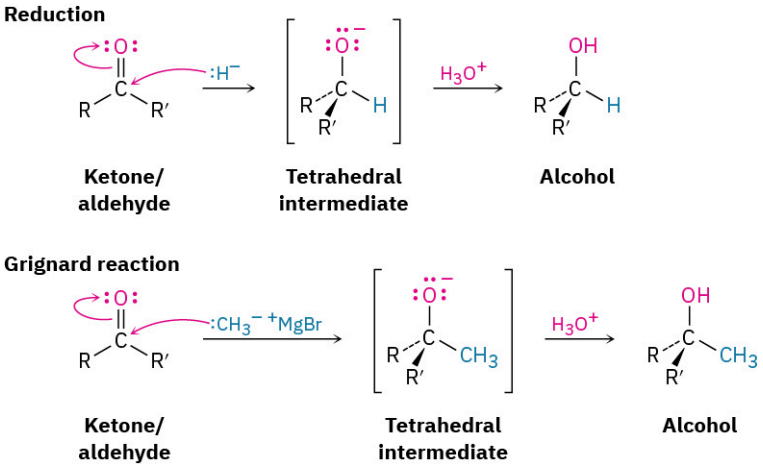


FIGURE 18.9 The addition reaction of an aldehyde or a ketone with a nucleophile. Depending on the nucleophile, either an alcohol or a compound with a C=Nu double bond is formed.

FORMATION OF AN ALCOHOL

The simplest reaction of a tetrahedral alkoxide intermediate is protonation to yield an alcohol. We've already seen two examples of this kind of process during reduction of aldehydes and ketones with hydride reagents such as $NaBH_4$ and $LiAlH_4$ (**Section 17.4**) and during Grignard reactions (**Section 17.5**). During a reduction, the nucleophile that adds to the carbonyl group is a hydride ion, $H{:}^-$, while during a Grignard reaction, the nucleophile is a carbanion, $R_3C{:}^-$.

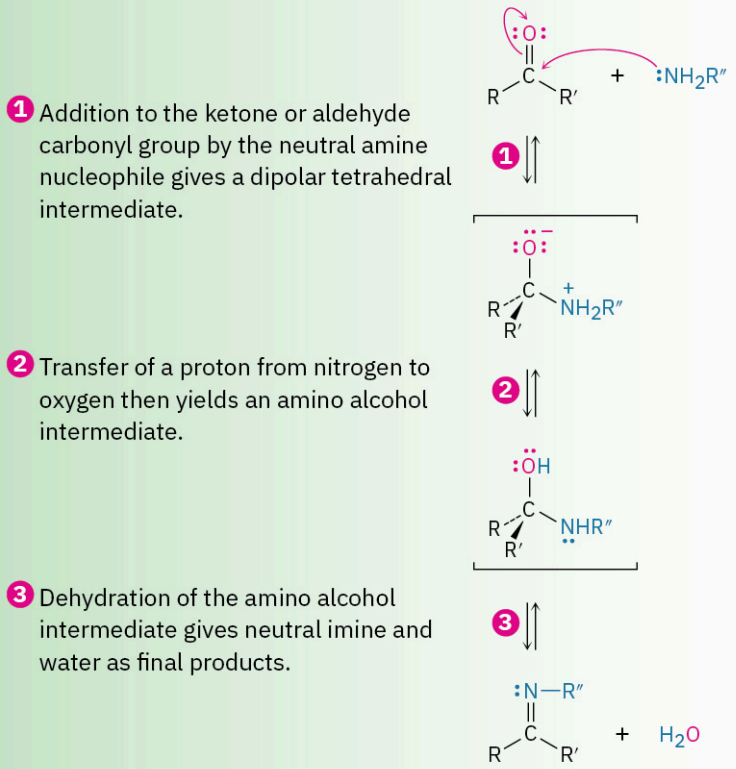


FORMATION OF C=Nu

The second mode of nucleophilic addition, which often occurs with amine nucleophiles, involves elimination of oxygen and formation of a C=Nu double bond. For example, aldehydes and ketones react with primary amines, RNH_2, to form *imines,* $R_2C{=}NR'$. These reactions use the same kind of tetrahedral intermediate as that formed during hydride reduction and Grignard reaction, but the initially formed alkoxide ion is not isolated. Instead, it is protonated and then loses water to form an imine, as shown in **FIGURE 18.10**.

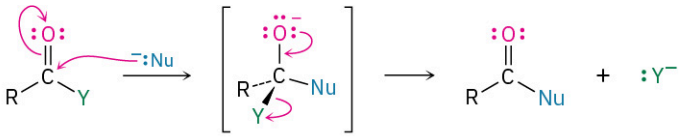
FIGURE 18.10 MECHANISM

Formation of an imine, $R_2C{=}NR'$, by reaction of an amine with an aldehyde or a ketone.



Nucleophilic Acyl Substitution Reactions of Carboxylic Acid Derivatives (Chapter 21)

The second fundamental reaction of carbonyl compounds, **nucleophilic acyl substitution**, is related to the nucleophilic addition reaction just discussed but occurs only with carboxylic acid derivatives rather than with aldehydes and ketones. When the carbonyl group of a carboxylic acid derivative reacts with a nucleophile, addition occurs in the usual way, but the initially formed tetrahedral alkoxide intermediate is not isolated. Because carboxylic acid derivatives have a leaving group bonded to the carbonyl-group carbon, the tetrahedral intermediate can react further by expelling the leaving group and forming a new carbonyl compound:



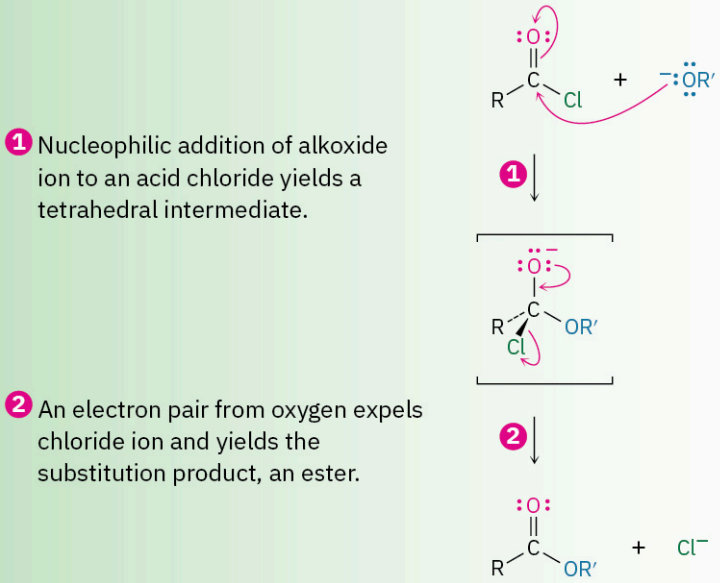
Carboxylic acid derivative **Tetrahedral intermediate**

[Y = –OR (ester), –Cl (acid chloride), $–NH_2$ (amide), or –OCOR (acid anhydride)]

The net effect of nucleophilic acyl substitution is the replacement of the leaving group by the entering nucleophile. We'll see in Chapter 21, for instance, that acid chlorides are rapidly converted into esters by treatment with alkoxide ions (FIGURE 18.11).

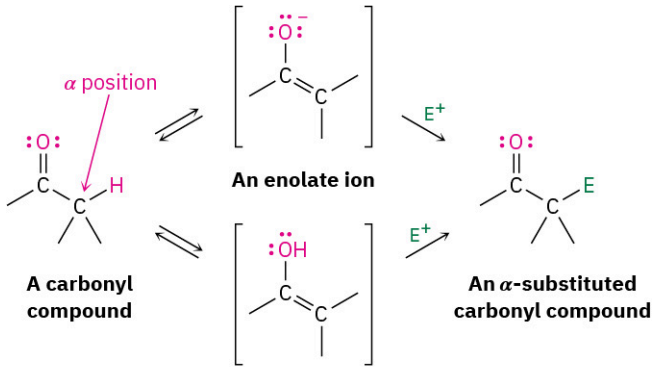
FIGURE 18.11 MECHANISM

Nucleophilic acyl substitution of an acid chloride with an alkoxide ion yields an ester.

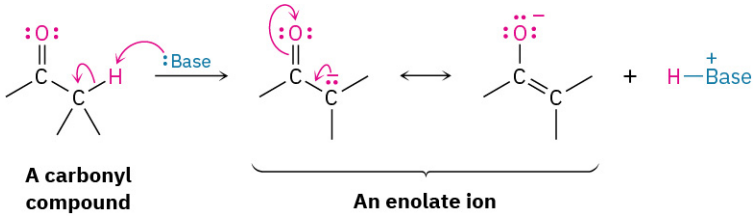


Alpha-Substitution Reactions (Chapter 22)

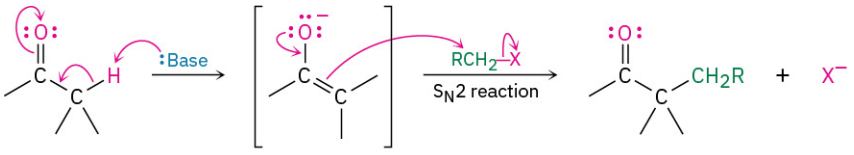
The third major reaction of carbonyl compounds, **alpha substitution**, occurs at the position next to the carbonyl group—the alpha (α) position. This reaction results in the substitution of an α hydrogen by an electrophile through the formation of an intermediate *enol* or *enolate ion:*



For reasons that we'll explore in Chapter 22, the presence of a carbonyl group renders the hydrogens on the α carbon acidic. Carbonyl compounds therefore react with strong base to yield enolate ions.



Because they're negatively charged, enolate ions act as nucleophiles and undergo many of the reactions we've already studied. For example, enolates react with primary alkyl halides in the S_N2 reaction. The nucleophilic enolate ion displaces halide ion, and a new C–C bond forms:

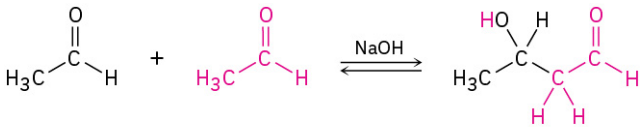


A carbonyl compound **An enolate ion**

The S_N2 alkylation reaction between an enolate ion and an alkyl halide is a powerful method for making C–C bonds, thereby building up larger molecules from smaller precursors. We'll study the alkylation of many kinds of carbonyl compounds in Chapter 22.

Carbonyl Condensation Reactions (Chapter 23)

The fourth and last fundamental reaction of carbonyl groups, **carbonyl condensation**, takes place when two carbonyl compounds react with each other. When acetaldehyde is treated with base, for instance, two molecules combine to yield the hydroxy aldehyde product known as *aldol* (*ald*ehyde + alcoh*ol*):

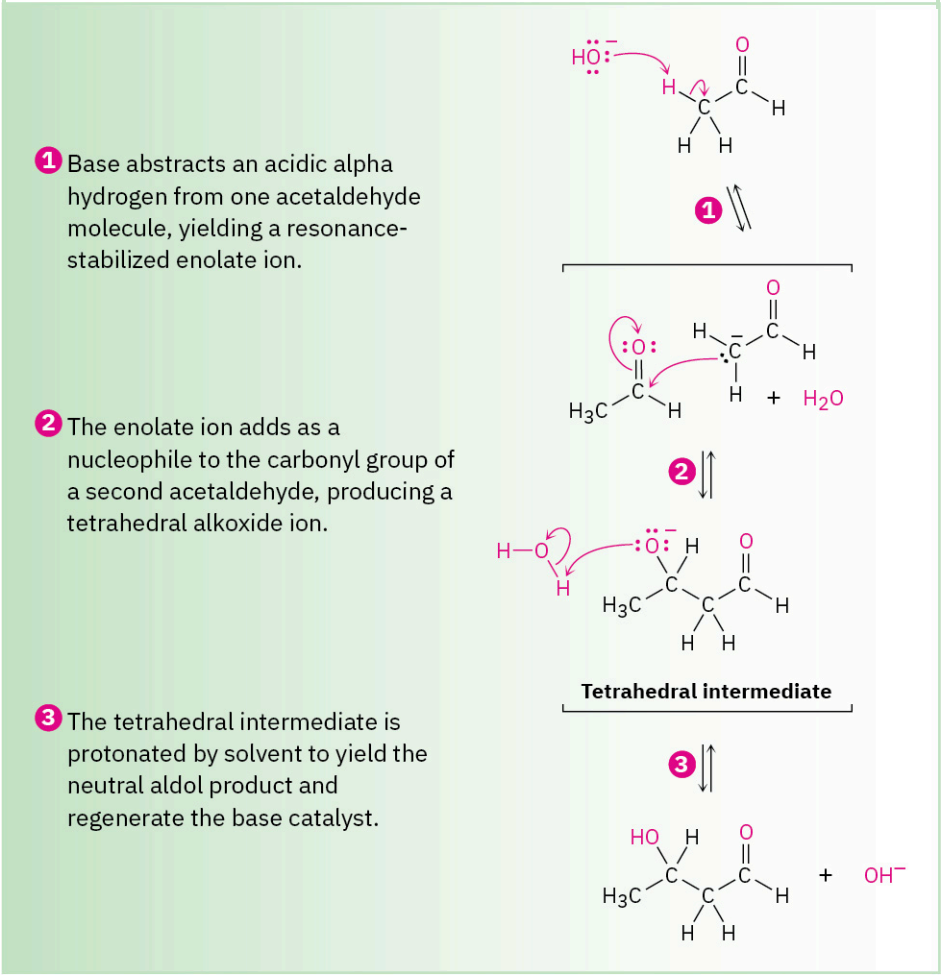


Two acetaldehydes **Aldol**

Although carbonyl condensation appears to be different from the three processes already discussed, it's actually quite similar. A carbonyl condensation reaction is simply a combination of a nucleophilic addition step and an α-substitution step. The initially formed enolate ion of one acetaldehyde molecule acts as a nucleophile and adds to the carbonyl group of another acetaldehyde molecule, as shown in FIGURE 18.12.

FIGURE 18.12 MECHANISM

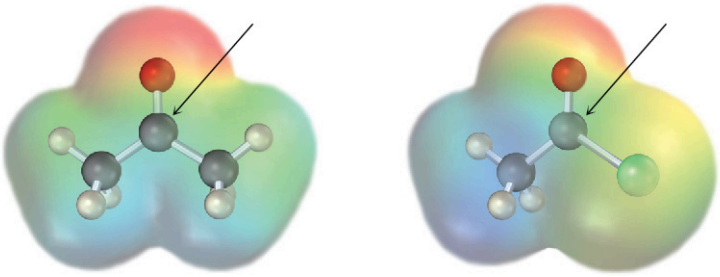
A carbonyl condensation reaction between two molecules of acetaldehyde yields a hydroxy aldehyde product.



IV Summary

To a great extent, the chemistry of living organisms is the chemistry of carbonyl compounds. We have not looked at the details of specific carbonyl reactions in this short preview but rather have laid the groundwork for the next five chapters. All the carbonyl-group reactions we'll be studying in Chapters 19 through 23 fall into one of the four fundamental categories discussed in this preview. Knowing where we'll be heading should help you keep matters straight in understanding this most important of all functional groups.

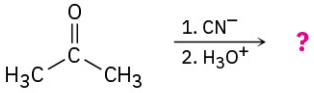
PROBLEM 18-69 Judging from the following electrostatic potential maps, which kind of carbonyl compound has the more electrophilic carbonyl carbon atom, a ketone or an acid chloride? Which has the more nucleophilic carbonyl oxygen atom? Explain.



Acetone
(ketone)

Acetyl chloride
(acid chloride)

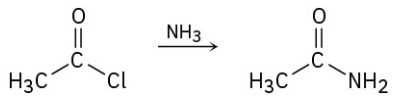
PROBLEM 18-70 Predict the product formed by nucleophilic addition of cyanide ion (CN^-) to the carbonyl group of acetone, followed by protonation to give an alcohol:



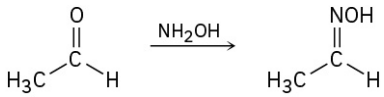
Acetone

PROBLEM 18-71 Identify each of the following reactions as a nucleophilic addition, nucleophilic acyl substitution, an α substitution, or a carbonyl condensation:

(a)



(b)



(c)

